# Transport and Mobility Modelling: Transport Demand (Week 2 of 2)

Anders Johansson

# Demand modelling: The Four Step Model

**Prerequisites:**
- Tessellation / discretisation of space (e.g. postcodes, wards, boroughs, countries)
- Population / demographic data

**Step 1: Trip generation:** How many trips originate *O* (e.g. residential areas) and destinate *D* (e.g. industrial/business areas) in each zone. This can be estimated based on zonal attributes (e.g. population, demographic, land use)

**Step 2:** Trip distribution: Estimation of origin-destination pairings.

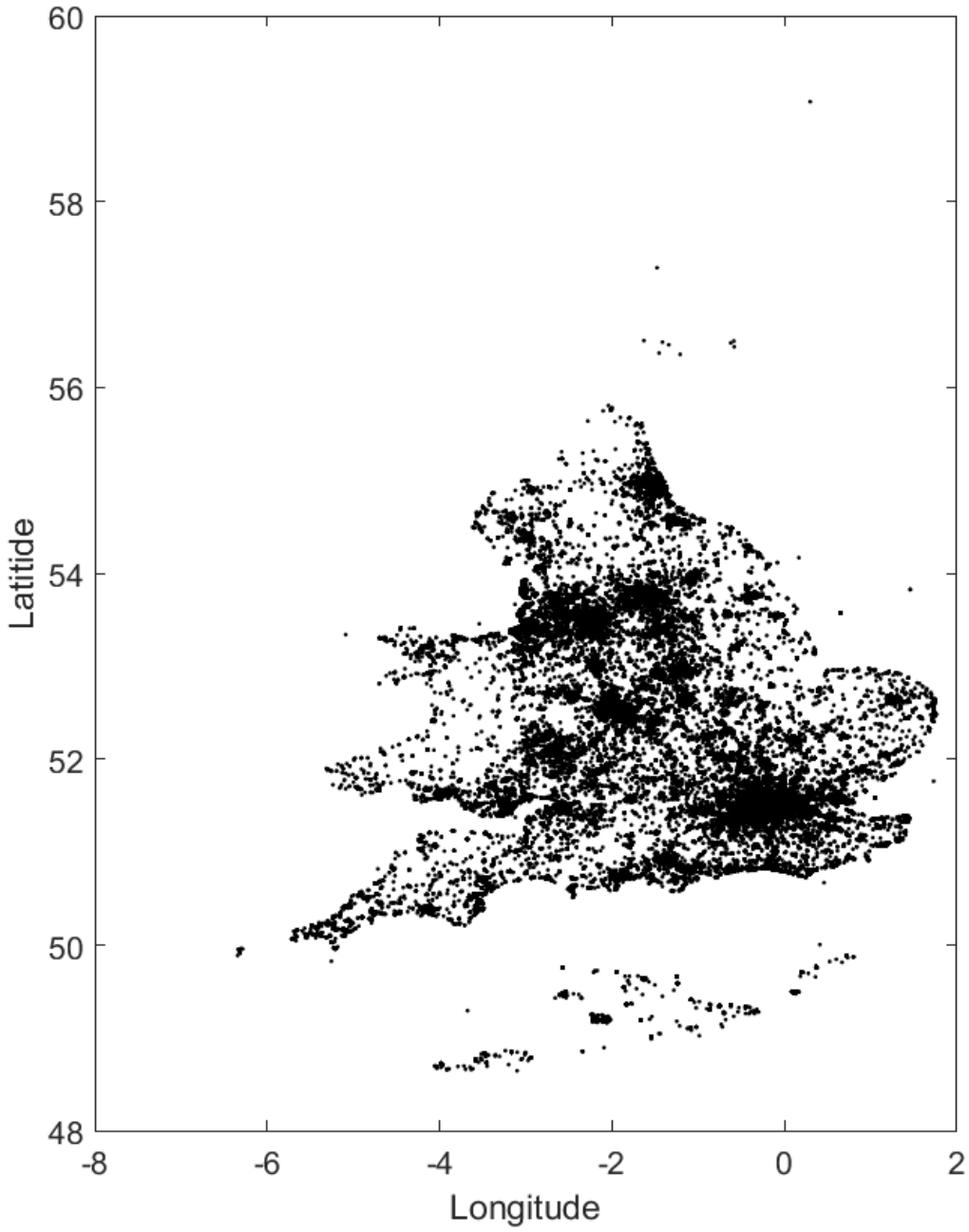
**Step 3: Mode choice / modal split** (walking, cycling, public transport, driving, rail, ...)

**Step 4: Traffic assignment:** Route choice.

# 6. Distance

# Distance

- Plotting the geo-location (longitude/latitude) of each tweet.
- To study trips we need to be able to quantify and measure ‘distance’.
- **But how do we measure distances between any two points on this map?**



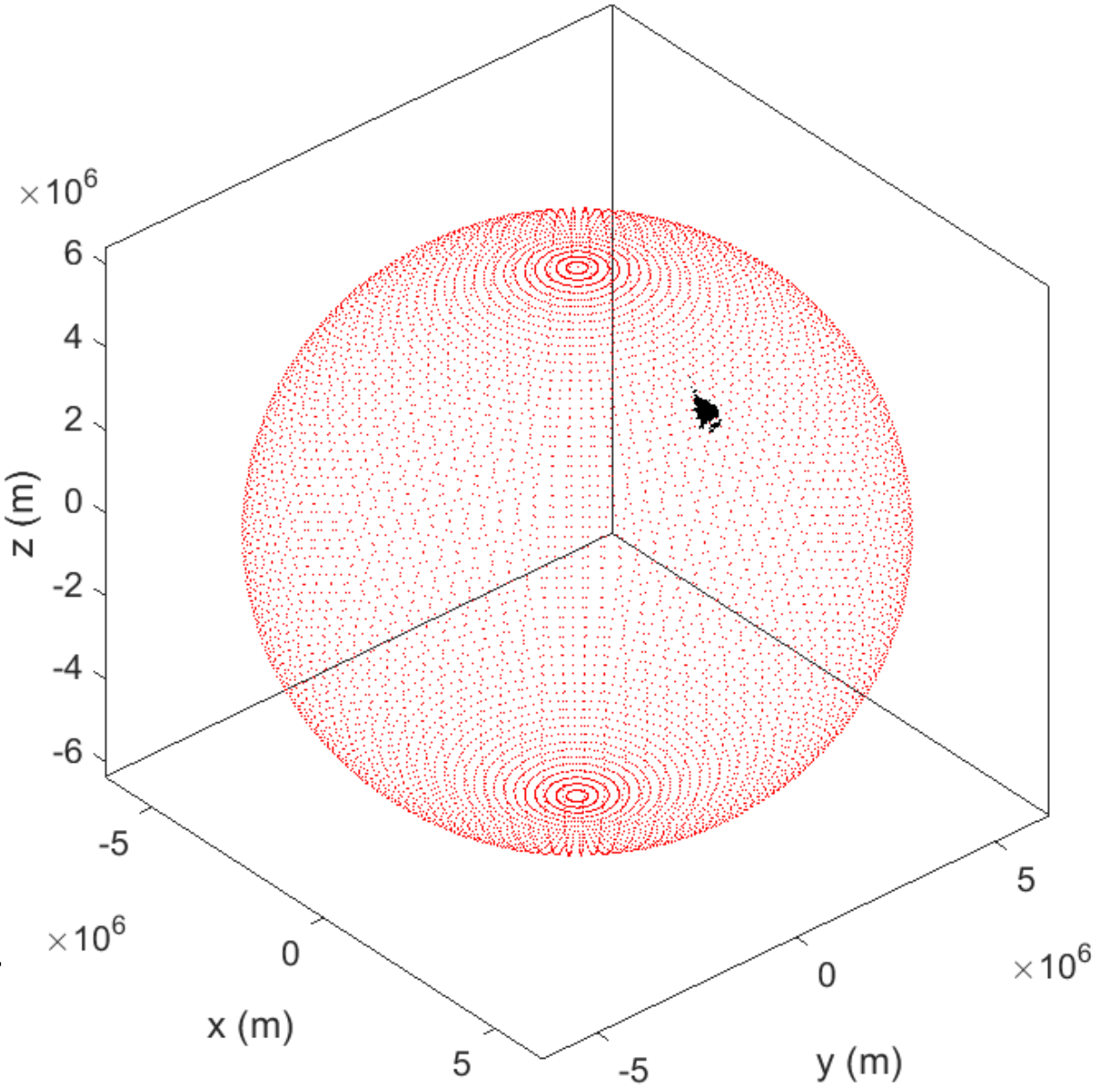
# Measuring distance

- If we approximate planet earth by a sphere of radius *r*=6,371 km, we can easily map all longitude/latitude pairs onto x, y, z space:

$$x = r\,cos(\theta)cos(\varphi)$$
$$y = r\,cos(\theta)sin(\varphi)$$
$$z = r\,sin(\theta)$$

- Then it is trivial to measure the distance between any two points in x, y, z space as the Euclidean distance:

$$\sqrt{(x_0 - x_1)^2 + (y_0 - y_1)^2 + (z_0 - z_1)^2}$$



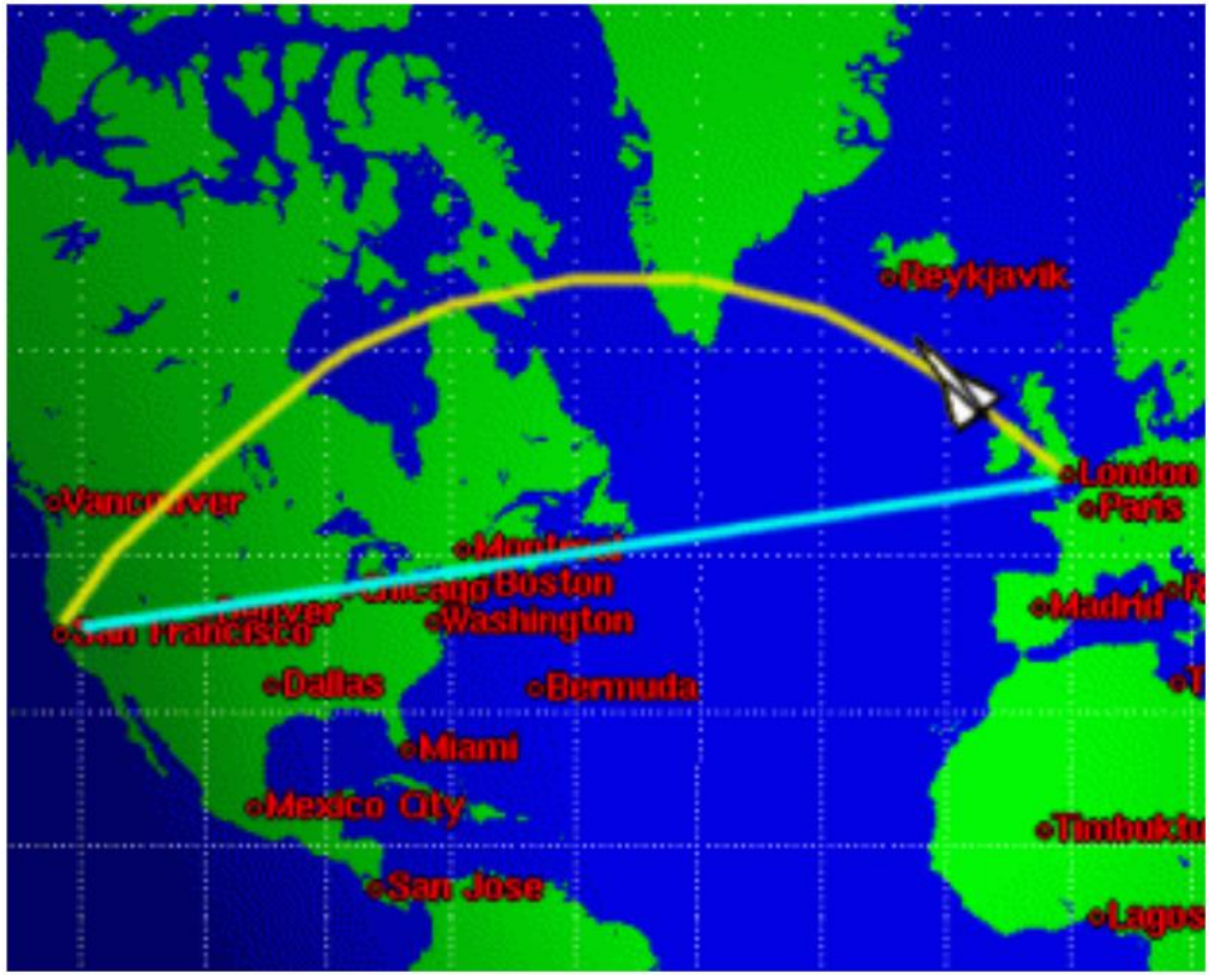
# Validating distance measurements

- In order to validate our distance estimation, we can test with a few known distances, e.g. City of London to Manchester (which should be around 262 km):

  *Our spherical approximation gives a distance of 267 km.* ✓

# Measuring distance

- In reality however the shortest distance between two points on a surface of a sphere is the *Geodesic* distance and not the *Euclidean* distance.



# Measuring distance

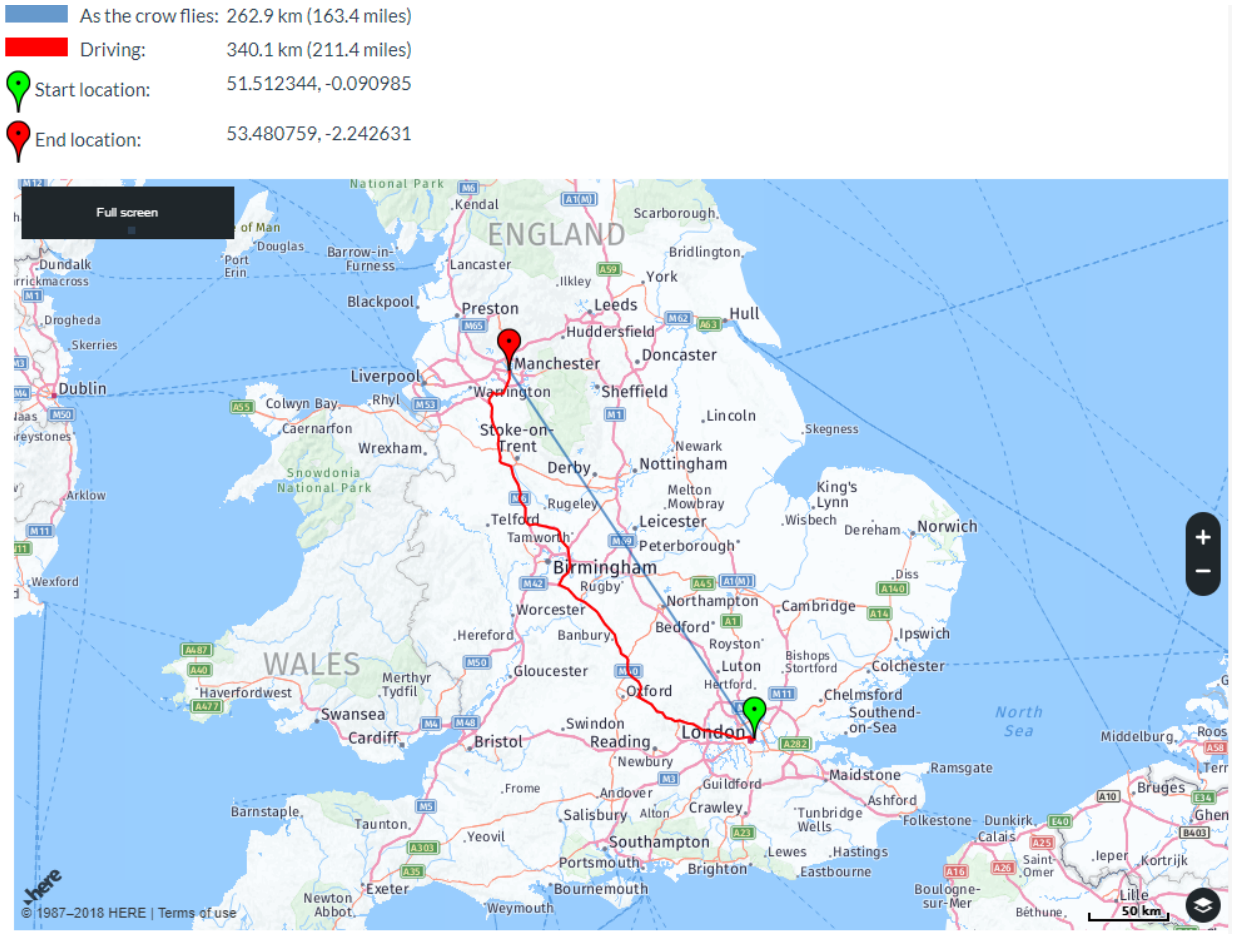
More importantly though is the notion of *what* the ‘distance’ really represent in our problem space. Is it:

- The Geodesic/Euclidean distance?
- The shortest path (if you are confined to the road network)?
- The quickest path (if you take congestion and speed into account)?
- The cheapest path (if you take transport cost into account)?
- The topological distance (e.g. number of stops with a train)?
- Is it the ‘generalised cost’ taking all of the above (and more) into account?

# Validating distance measurements

- In order to validate our distance estimation, we can test with a few known distances, e.g. City of London to Manchester (which should be around 262 km):

  *Our spherical approximation gives a distance of 267 km. ✓*




https://www.doogal.co.uk/MeasureDistances.php

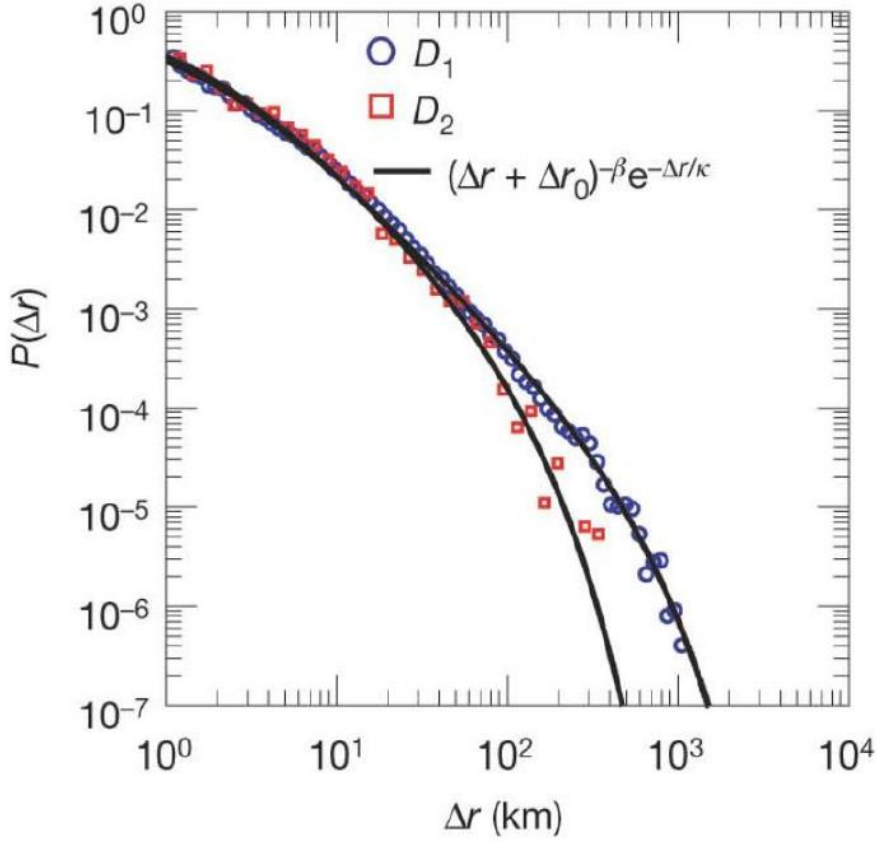
# Measuring distance

For the remainder of this lecture we will use the Euclidean distance to keep things simple.

# Distribution of trip lengths (literature)

..

Gonzalez et al., Understanding individual human mobility patterns, Nature, 2008.




D1: 100,000 for 6 months
D2: 200, 1 week, every 2h

$$\beta = 1.75 \pm 0.15$$

$$\Delta r_0 = 1.5km$$

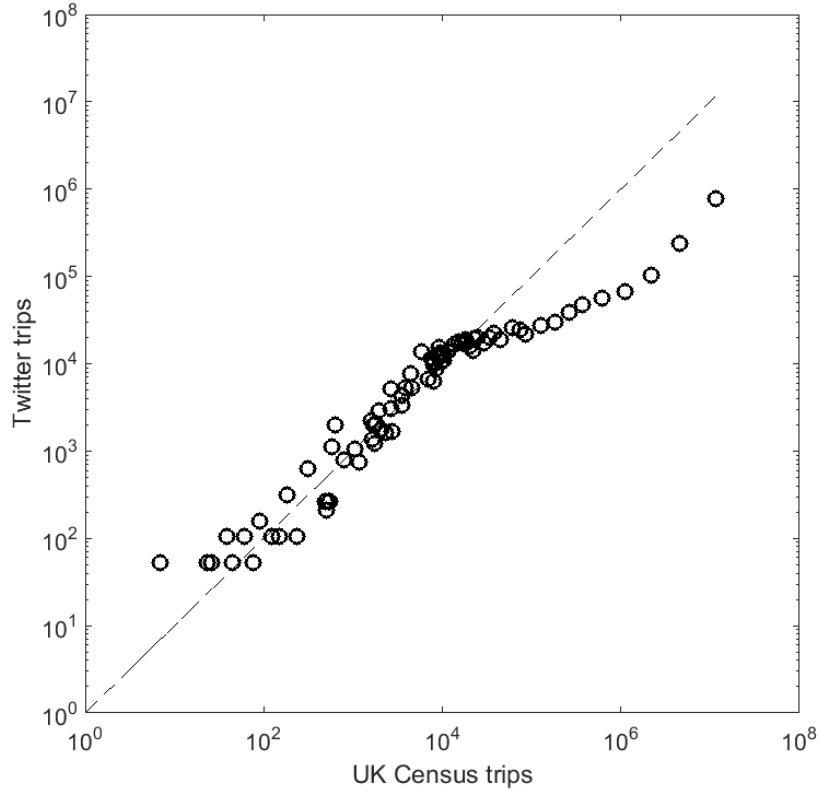
$$\kappa = 400km$$

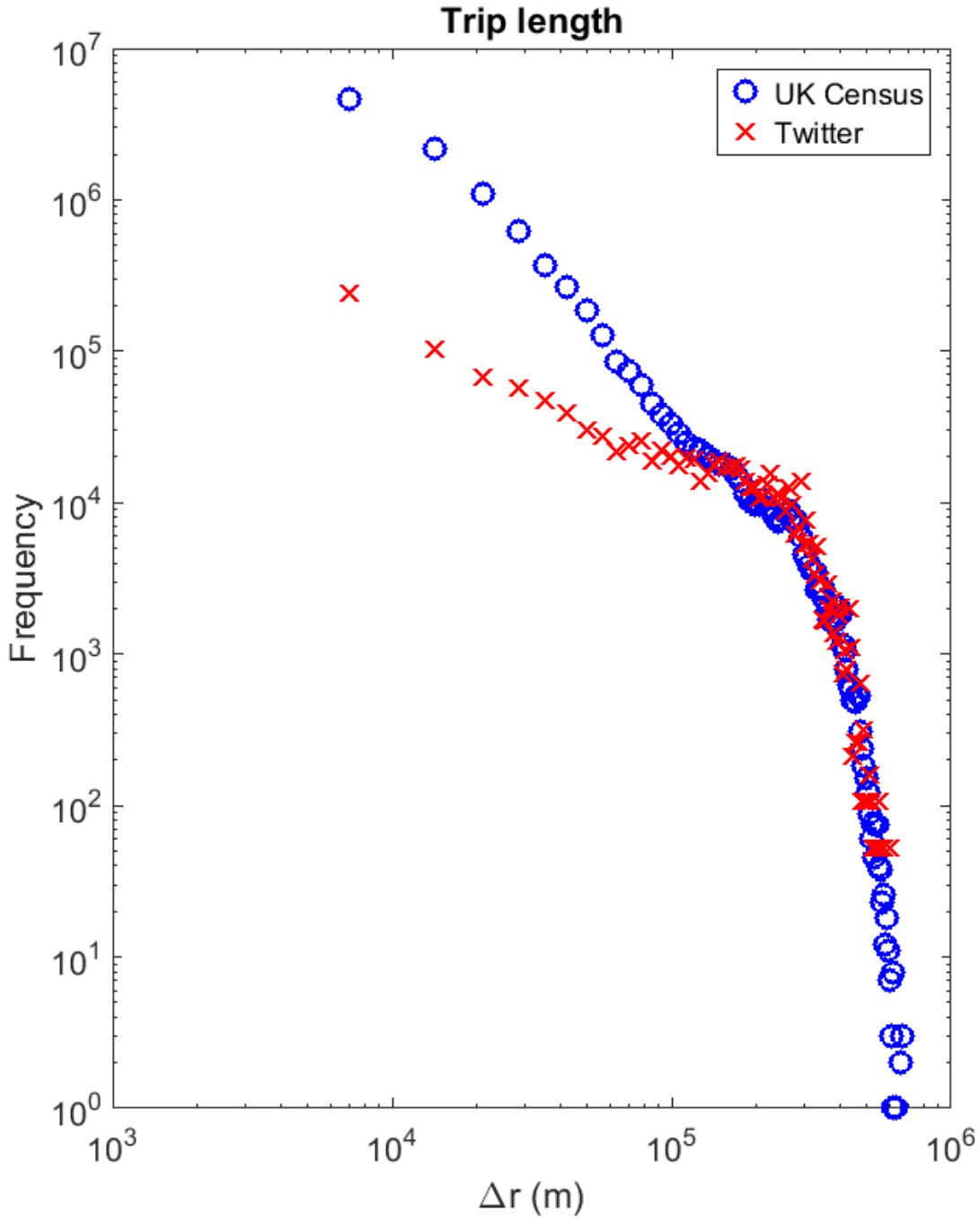
$$P(\Delta r) = (\Delta r + \Delta r_0)^{-\beta} \exp(-\Delta r/\kappa)$$

F Simini et al, Human mobility tutorial

# Distribution of trip lengths (our data)

- Similar at long distances but different for short distances.



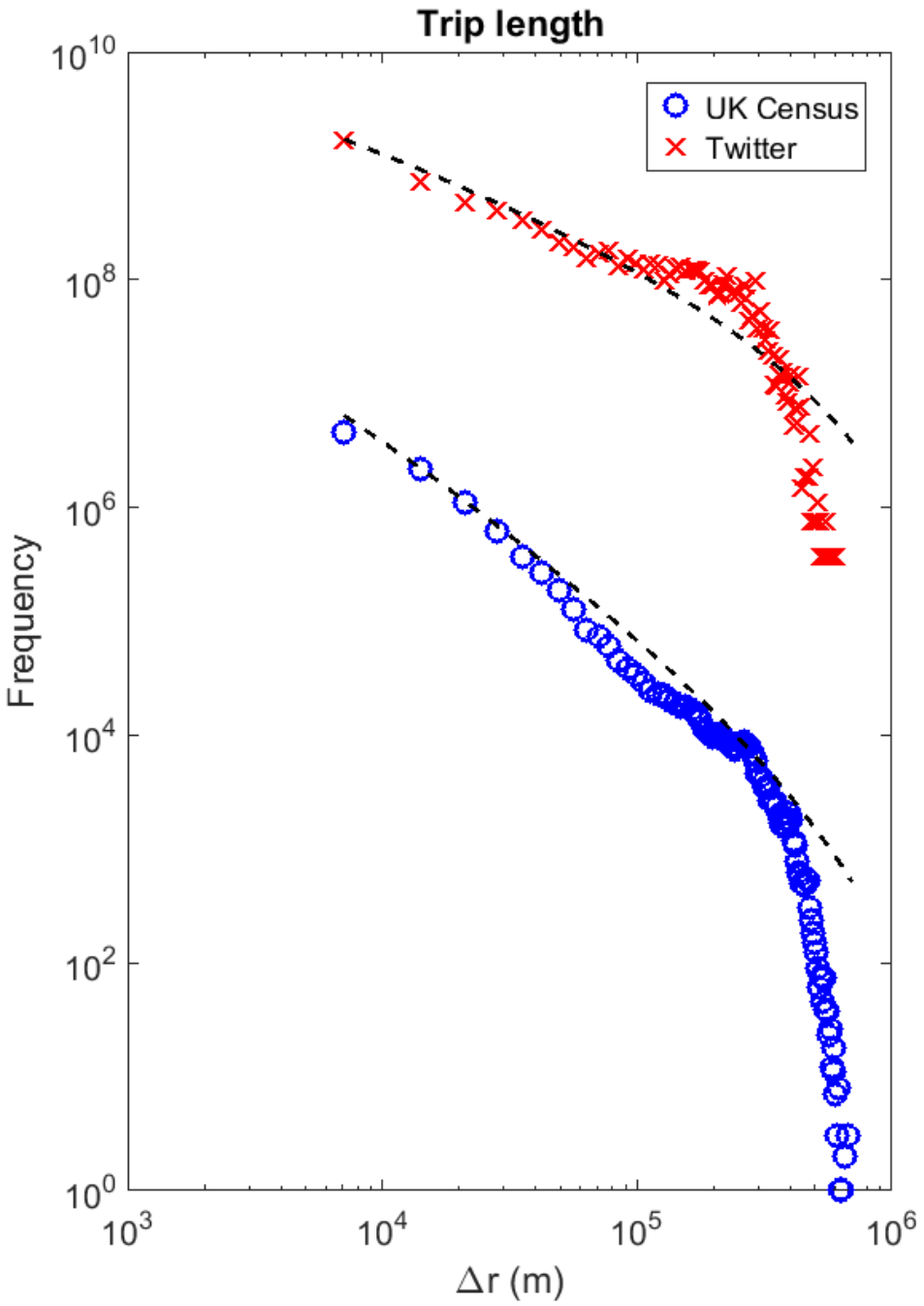
# Distribution of trip lengths (our data)

Similar to literature (US data), but two differences:

- There is a more rapid drop-off at distances around 300 km.
- The slope is slightly different

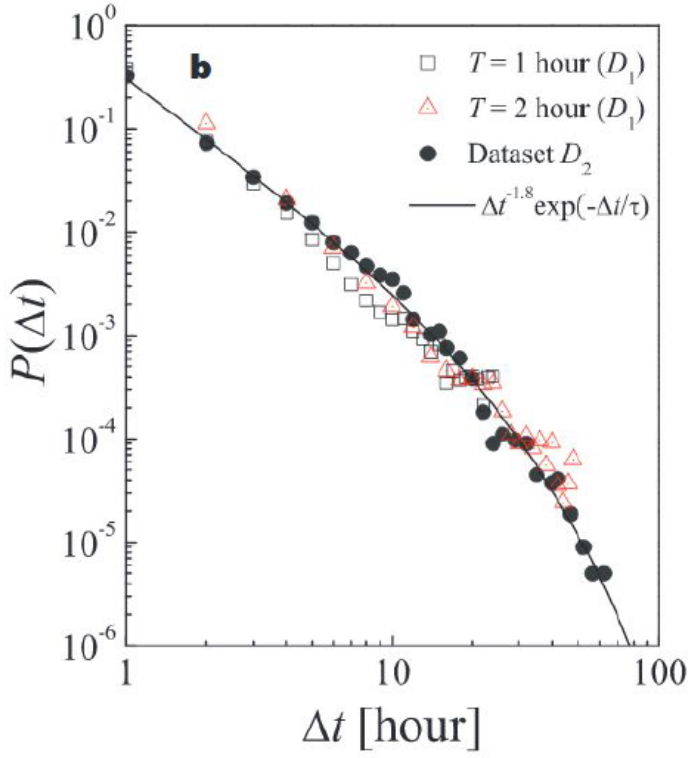
β = 1.75 (same as in literature) for UK Census trips

β = 1.25 for Twitter trips.



# Elapsed time between trips (literature)

Modelling the scaling properties of human mobility (Song et al., Nature Physics, 2010).




$$\beta_t = 0.8 \pm 0.1$$

$$\tau = 17h$$

$$P(\Delta t) = \Delta t^{-1-\beta_t} \exp(-\Delta t/\kappa)$$

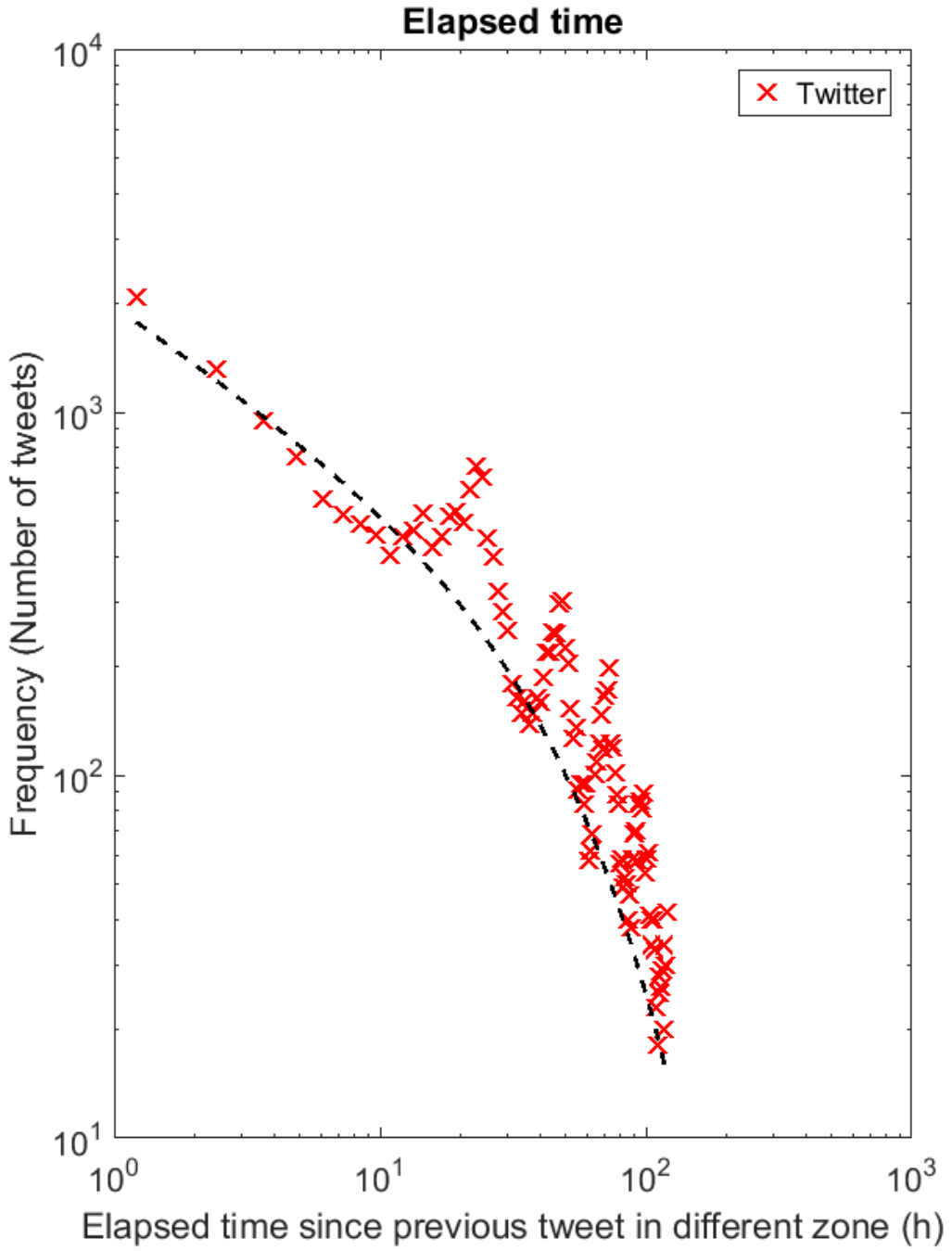
F Simini et al, Human mobility tutorial

# Elapsed time between trips (our data)

Compared with the literature, our data has got different fit function parameters:

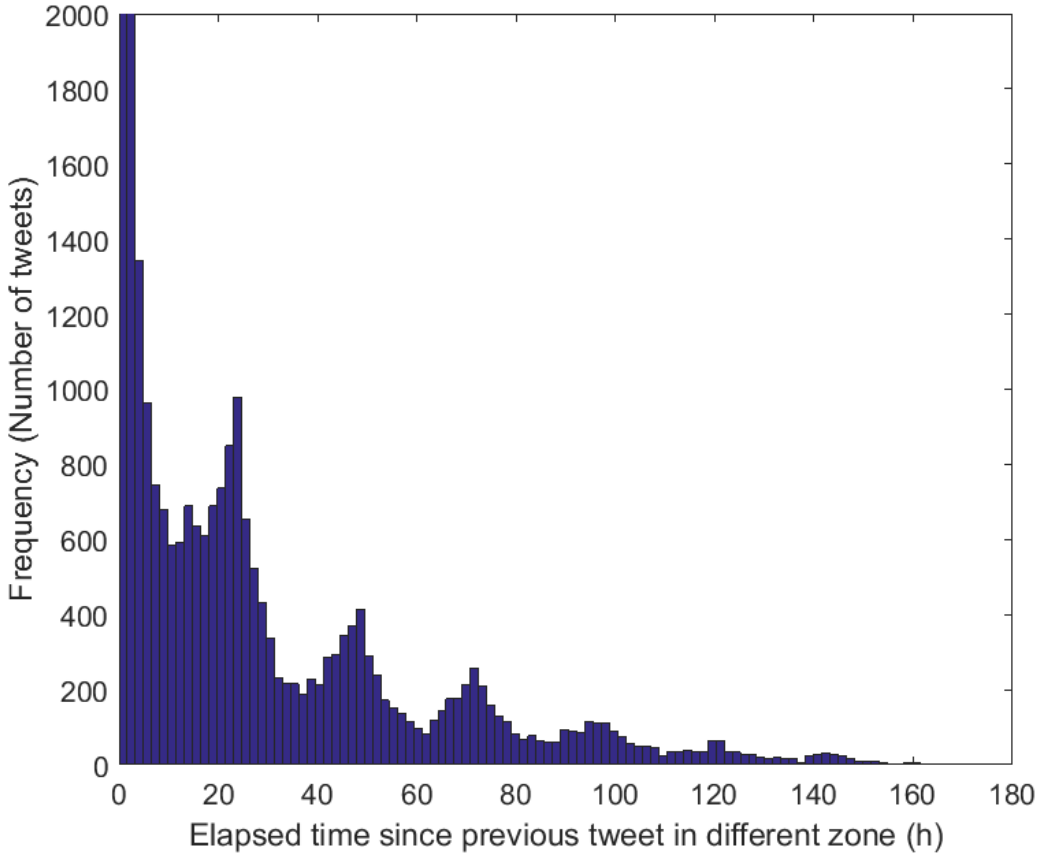
$\beta_t = -0.5$

$\tau = 48$ h



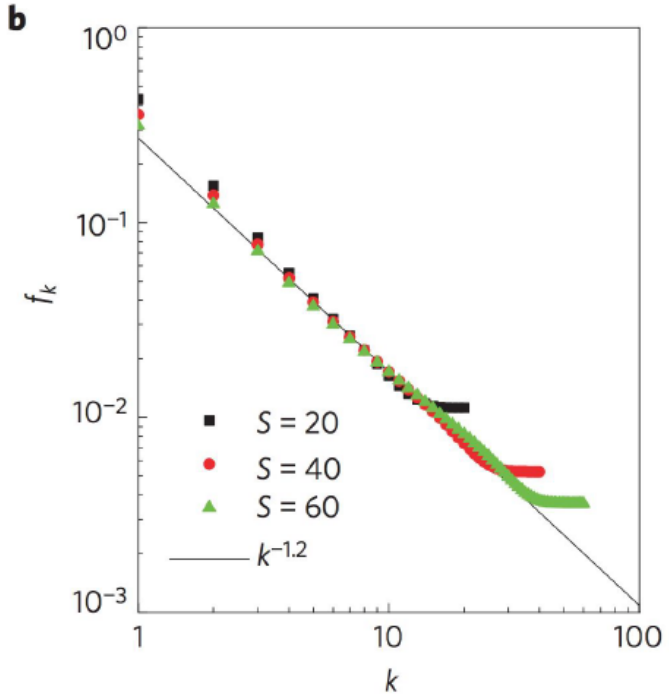
# Elapsed time between trips (our data)

There is also some periodicity in our Twitter dataset.



# Location-visitation frequency (literature)

The visitation frequency of the k-th most visited location of an individual




A Zipf's law:

$$f_k = k^{-\xi}$$

$$\xi = 1.2 \pm 0.1$$

Song et al., Modelling the scaling properties of human mobility, Nature Physics, 2010.

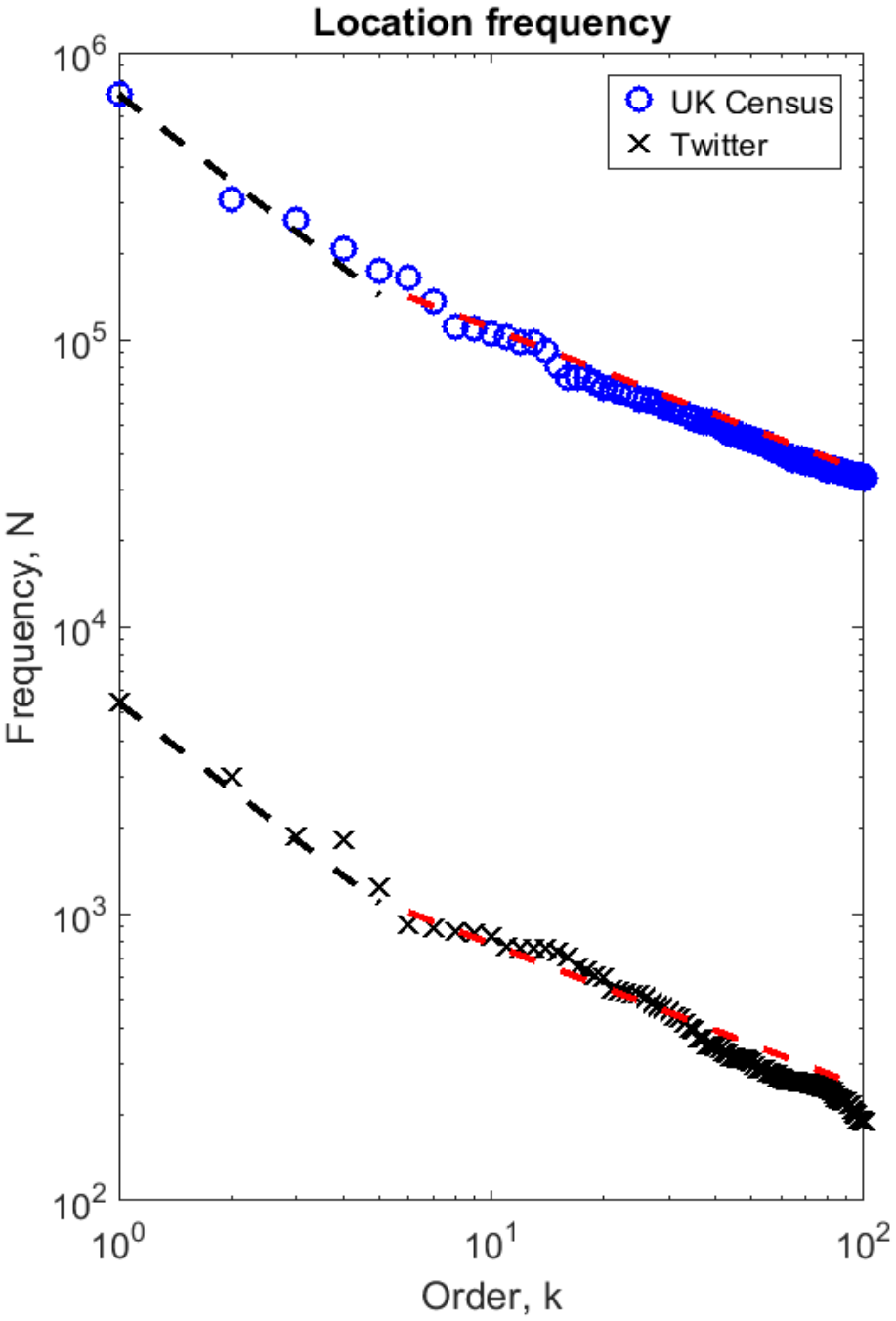
F Simini et al, Human mobility tutorial

# Location-visitation frequency (our data)

Similar trend, but the slope is different from literature (US data):

ξ = 1.0 for both UK Census D trips and Twitter frequency for the top 5 locations, and

ξ = 0.5 for locations 6 to 100.

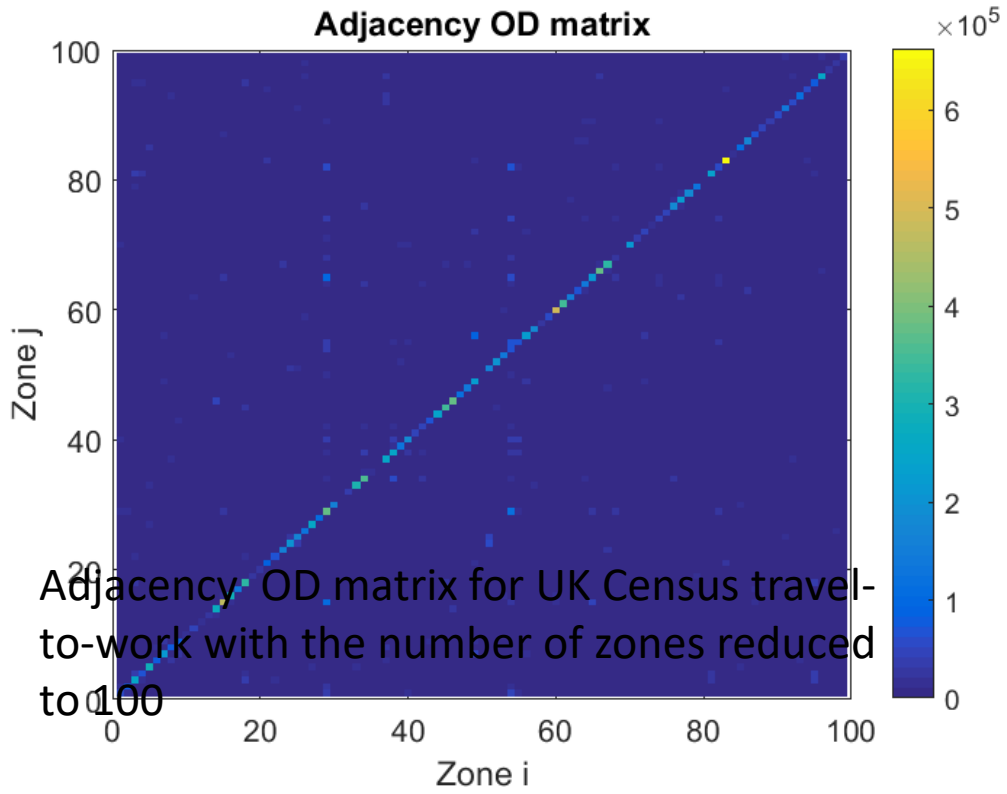


# 7. FLOW DATA, T_ij

# Flow data: OD matrix

Let us now turn our attention to trip estimation

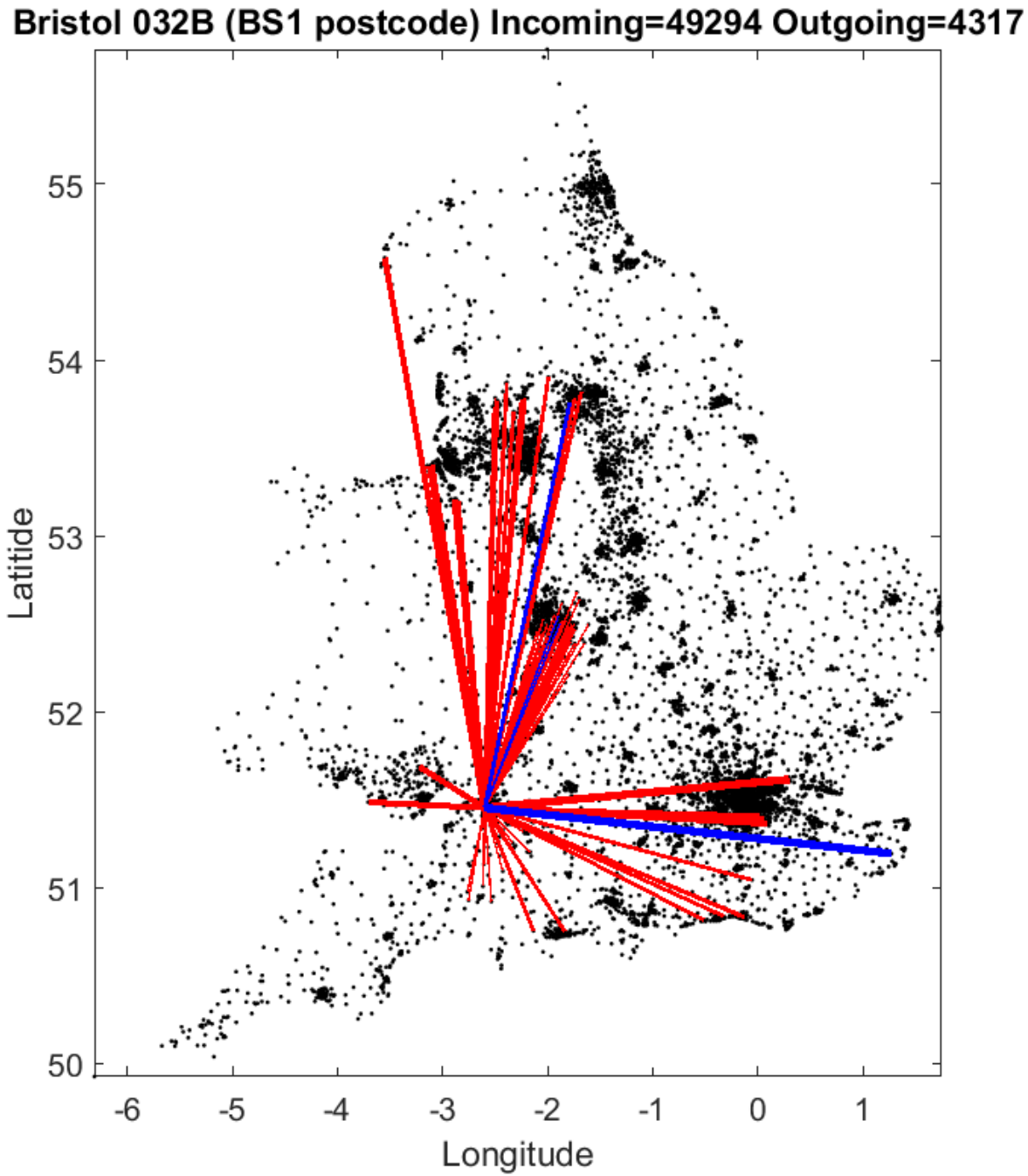
- $T_{ij}$ = number of trips from zone *i* to zone *j* per unit time
- There are a number of different models that can be used to estimate $T_{ij}$: The Gravity model, Radiation model, Entropy maximization, Intervening Opportunities model
- Also, the UK Census data provides data for travel-to-work journeys for all trips i to j, broken down in age groups and genders.




Adjacency OD matrix for UK Census travel-to-work with the number of zones reduced to 100

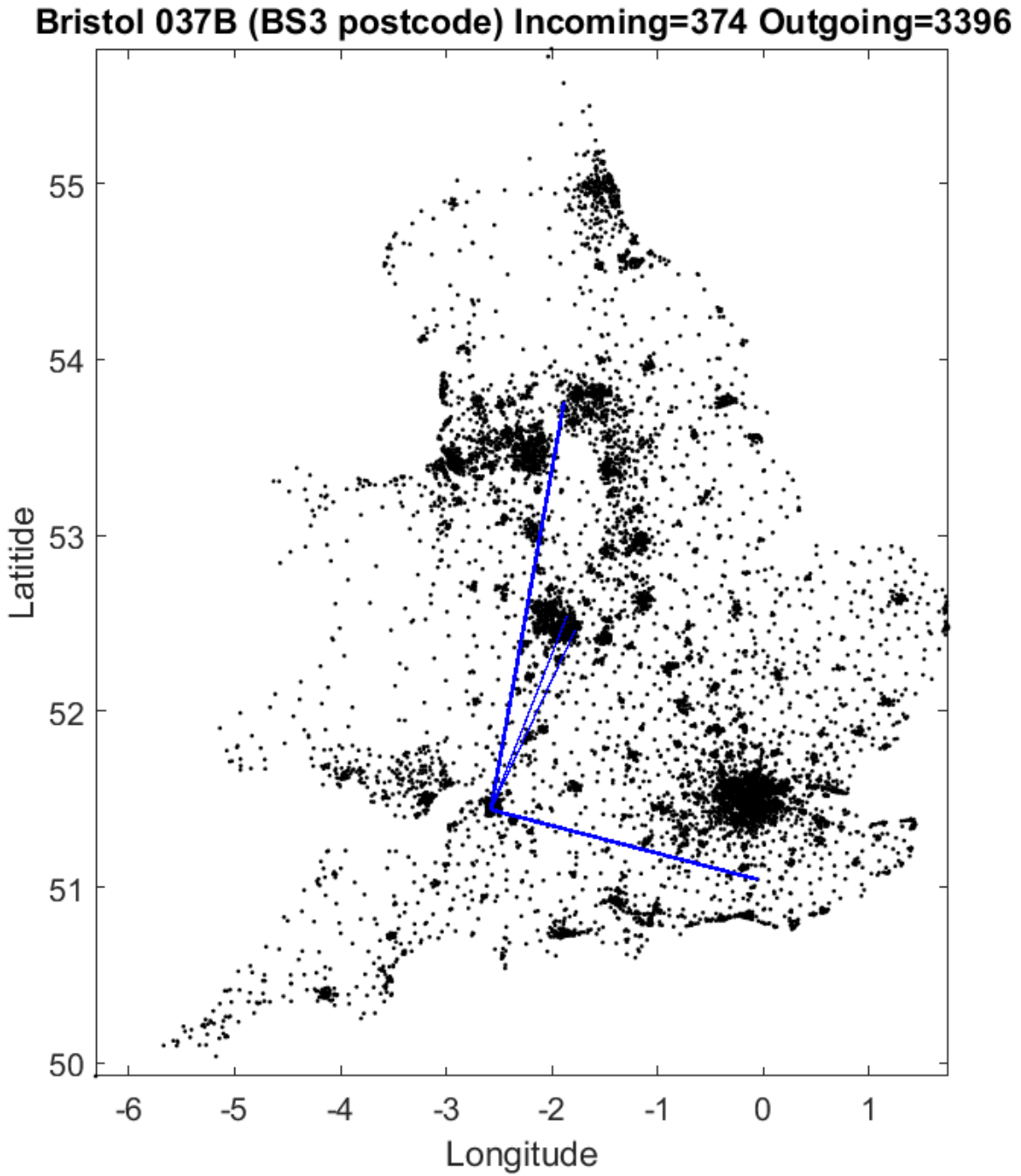
# Flow data: OD matrix

Largest flows in (red) and out (blue).



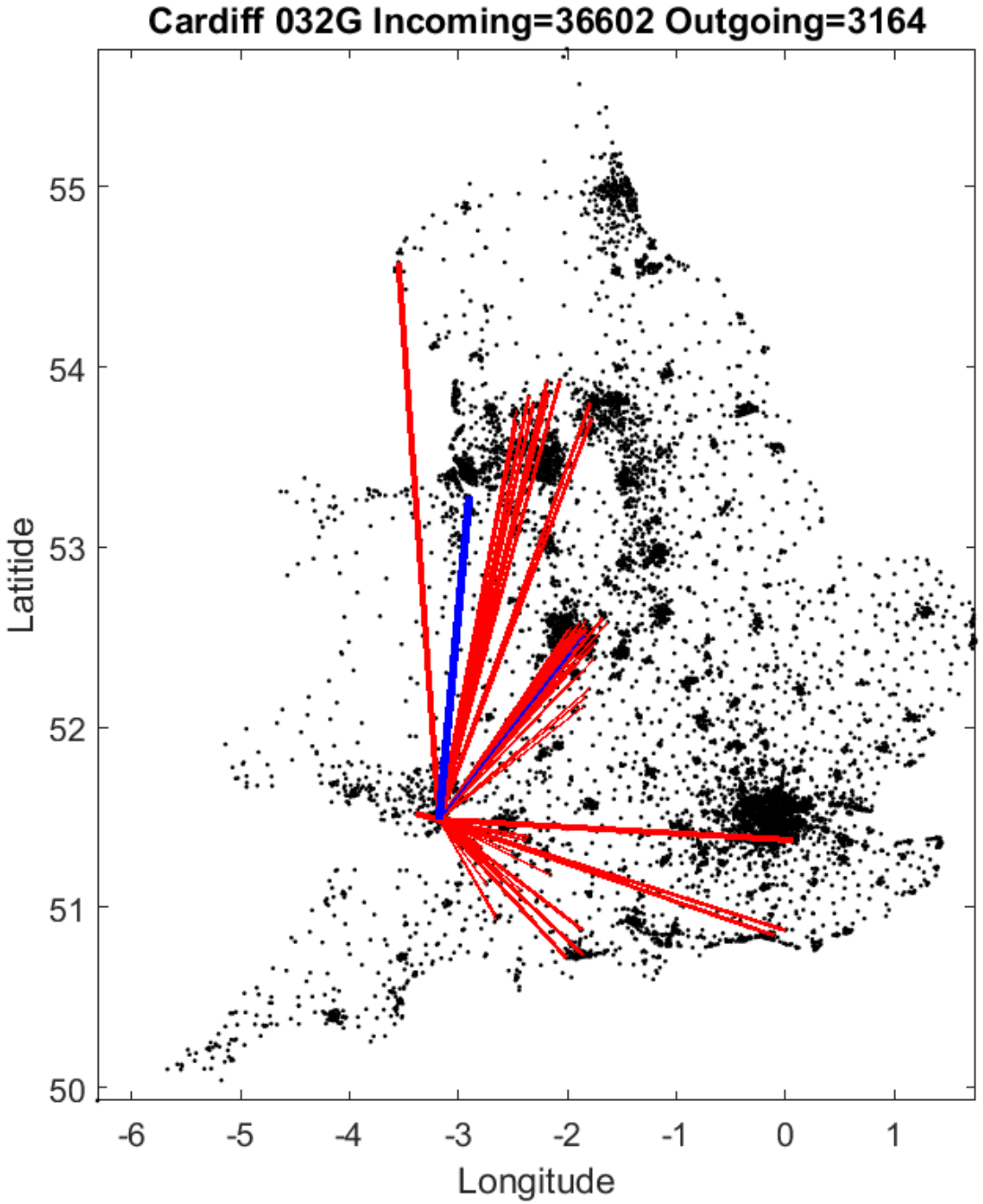
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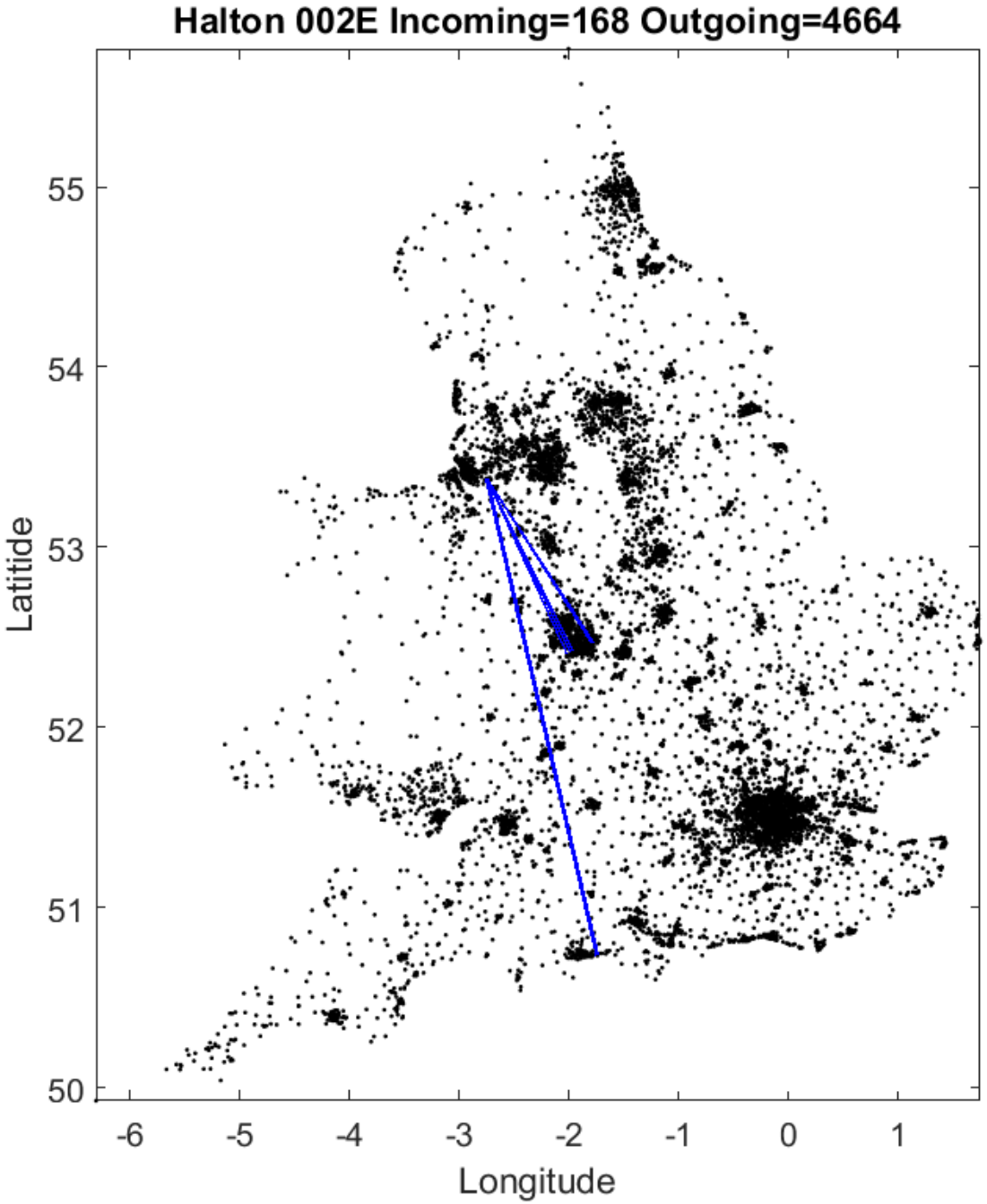
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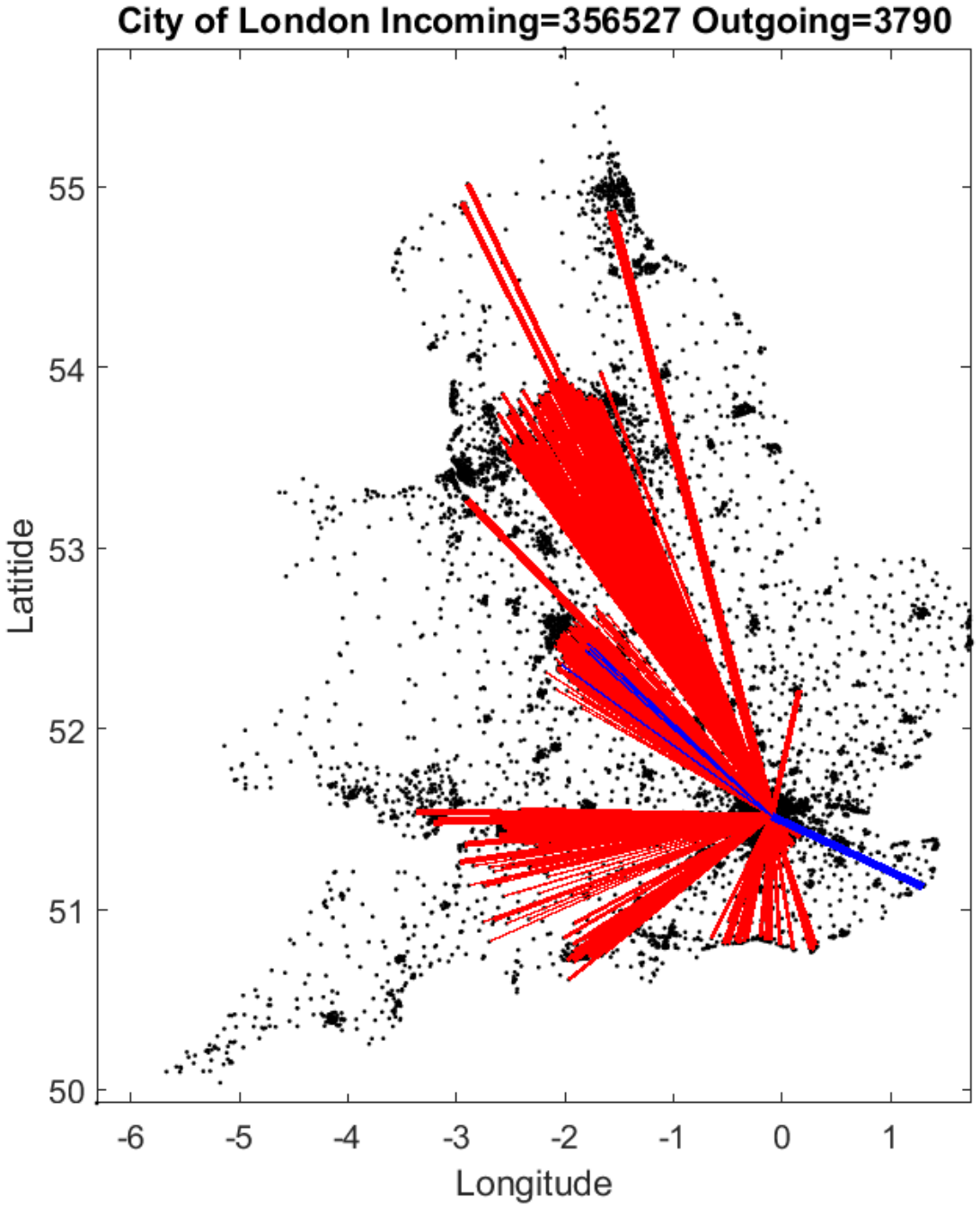
# Flow data: OD matrix

Largest flows in (red) and out (blue).



# Flow data: OD matrix

Largest flows in (red) and out (blue).



# Flow data: OD matrix

Let us now see if we can predict the number of trips from zone i to zone j.

# Demand modelling: Trip distribution

## THE $\frac{P_1 P_2}{D}$ HYPOTHESIS: ON THE INTERCITY MOVEMENT OF PERSONS

GEORGE KINGSLEY ZIPF
*Harvard University*

IN THE present paper we shall show with supporting data that the number of persons that move between any two communities in the United States whose respective populations are $P_1$ and $P_2$ and which are separated by the shortest transportation distance, $D$, will be proportionate to the ratio, $P_1 \cdot P_2/D$, subject to the effect of modifying factors.

The data in support of the above proposition are the highway, railway and airway data for an arbitrary set of cities during intervals of measurement in 1933-34. Before presenting the data, however, we shall give a brief theoretical discussion of the proposition itself with illustrations from other kinds of observations with which the above data are intimately connected.

### 1. THEORETICAL DISCUSSION

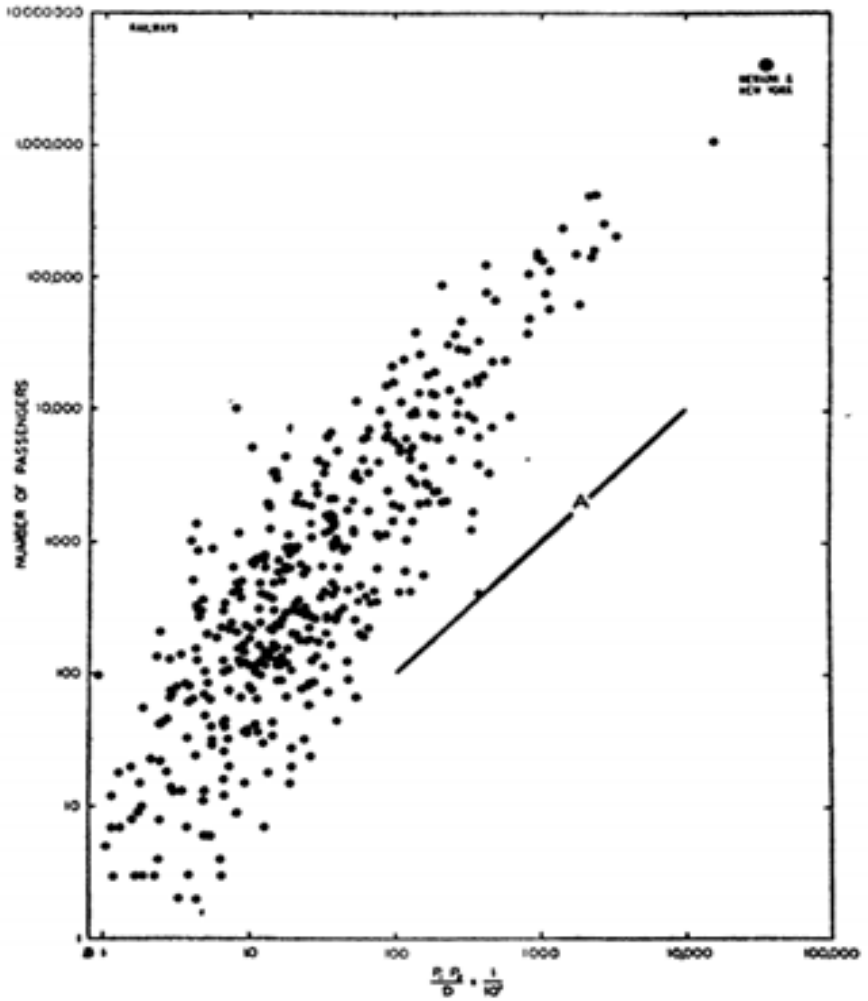
In 1940 the author published the observation that the following equation of the generalized harmonic series described the recent distribution of communities in India, Germany, and certain other countries including the United States (for communities of 2,500 or more inhabitants), when the communities are arranged in the order of decreasing size, with $A$ representing the population of the largest community, and with the denominators referring to the ranks of the communities thus arranged:[1]

$$A\ Sn = \frac{A}{1^p} + \frac{A}{2^p} + \frac{A}{3^p} + \ldots + \frac{A}{n^p}$$

In Figure 1 are presented the United States urban data for 1930 and 1940, as indicated, to which an ideal line, $A$, with a negative slope of 1 (i.e. $p = 1$) has been added to aid the reader's eye. The linearity of the data is apparent.

In 1941 the author presented a fuller

[1] Zipf, G. K. "The generalized harmonic series as a fundamental principle of social organization." *Psychological Record*, 4 (1940), 43.

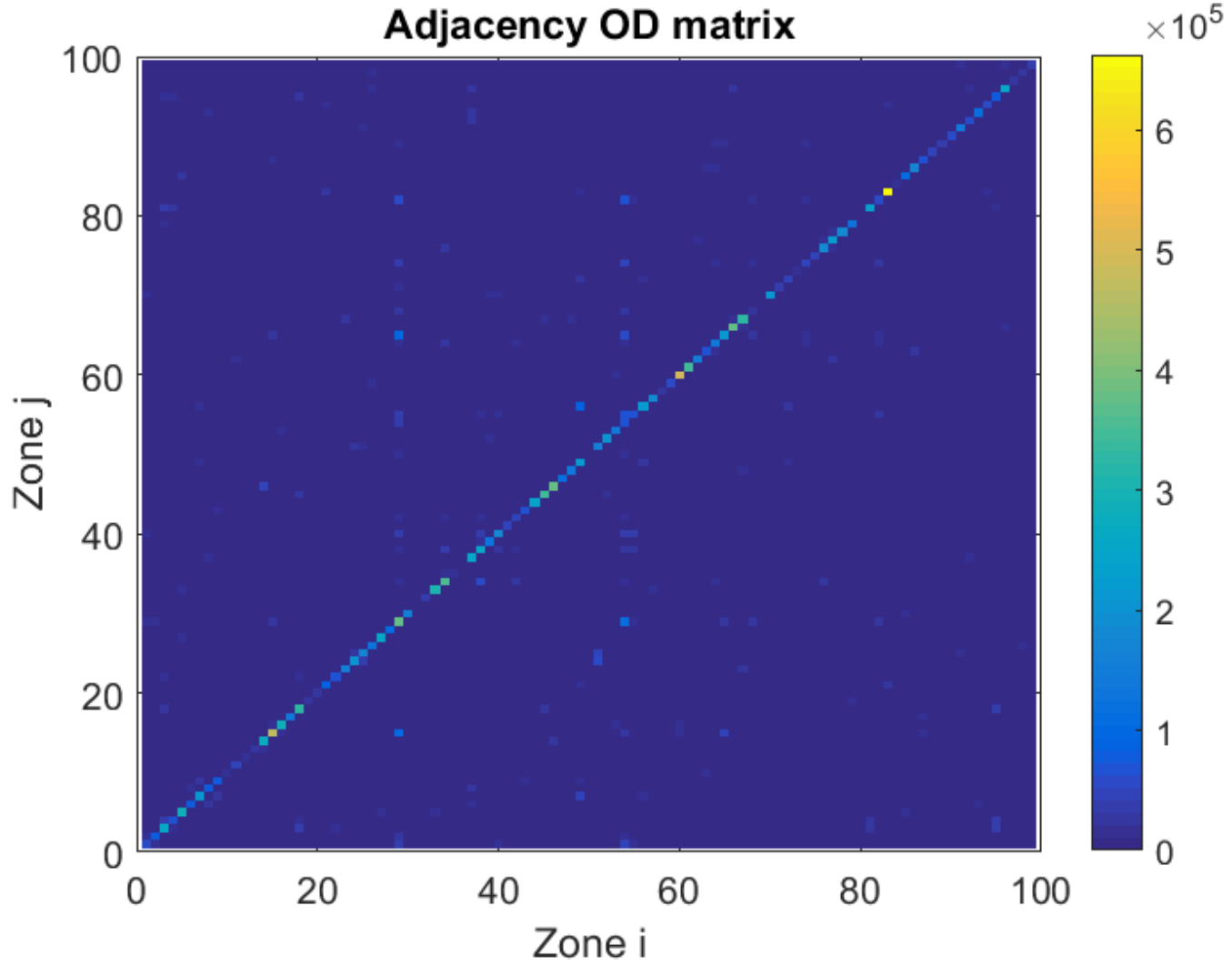


# Demand modelling: The Gravity model

Let us now try to predict the number of trips from zone *i* to zone *j* using the Gravity model*:*

$T_{ij} \propto P_i P_j f(d)$

where
- $P_i$ $(P_j)$ is the population in zone *i (j)*
- *f(d) = ??*



# Flow data: Functional form of f(d)

The function *f(d)* is a monotonically decreasing by increasing distance *d*. However many different specifications have been proposed in the literature, such as:

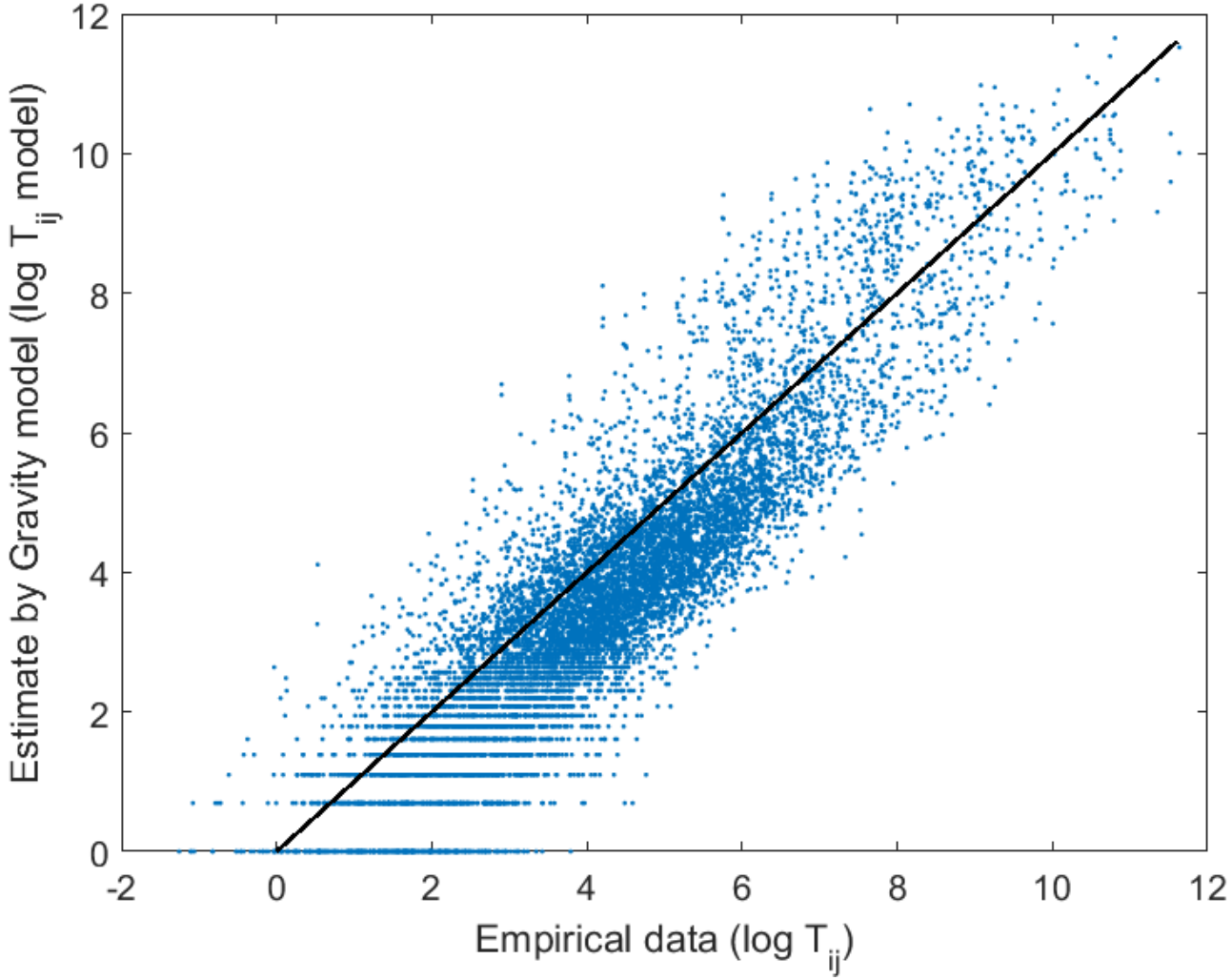
$f(d) = 1/d$
$f(d) = 1/d^2$
$f(d) = exp(-\beta d)$

# Flow data: Trips $T_{ij}$

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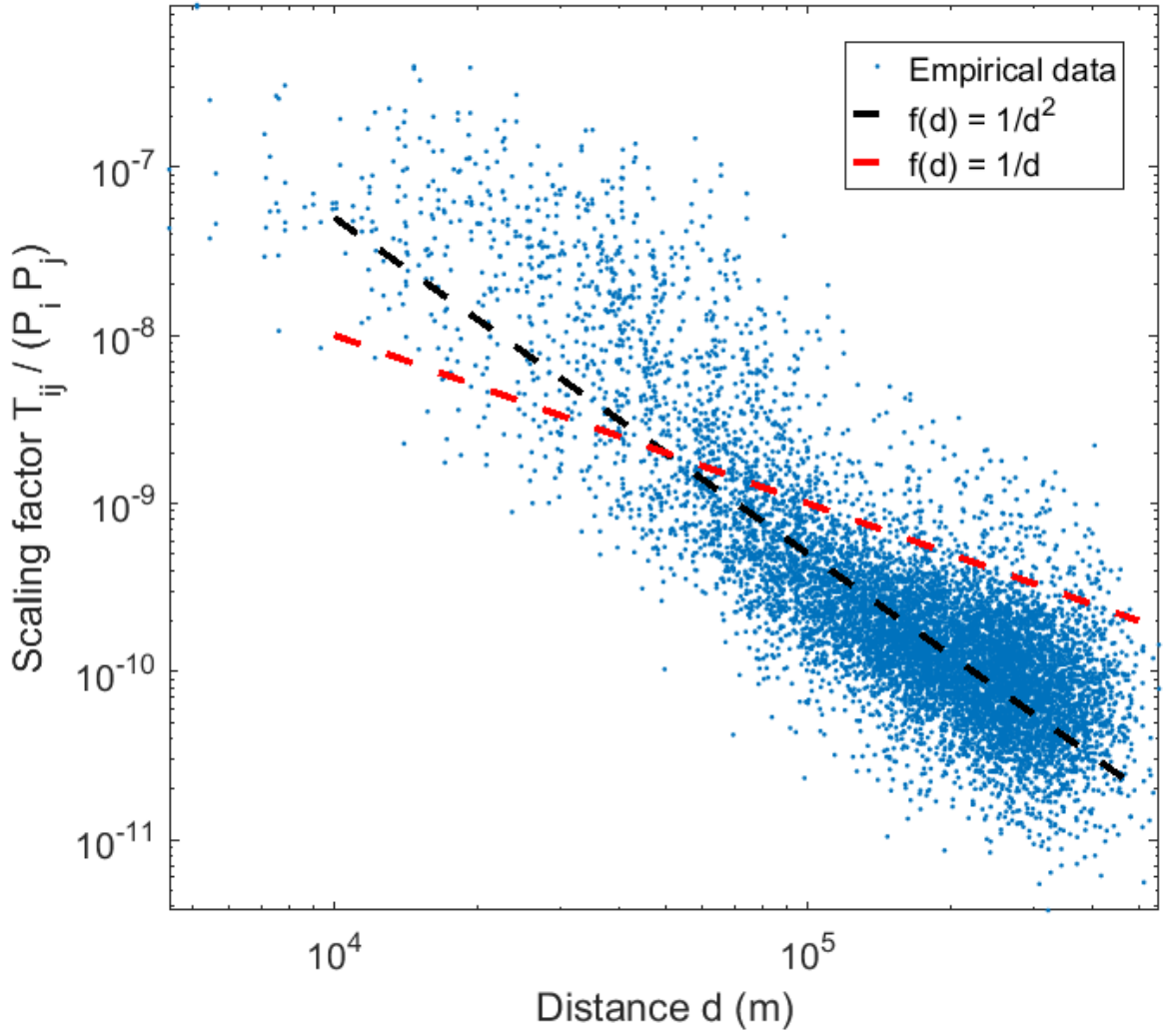
where
- $P_i$ $(P_j)$ is the population in zone *i (j)*
- $f(d) = 1/d^2$



# Flow data: Functional form of f(d)

To find out the best fit of f(d) to UK Census commute flows, let us do the following:

$f(d) \propto T_{ij} / (P_i P_j)$



# Constraints

The Gravity model accuracy can be improved if it is constrained:

- **Globally constrained**: The sum total generated trips matches the expected total number of trips.
- **Origin constrained**: Scaling coefficient to ensure that all the sum of all origin flows $O_i$ from $i$ correspond to empirical data.
- **Destination constrained**: Same as origin constrained but for destination flows $D_j$
- **Doubly constrained**: Both of the constraints above simultaneously.

# 8. Theory and models

# Demand modelling: Trip distribution

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### 1. THEORETICAL DISCUSSION

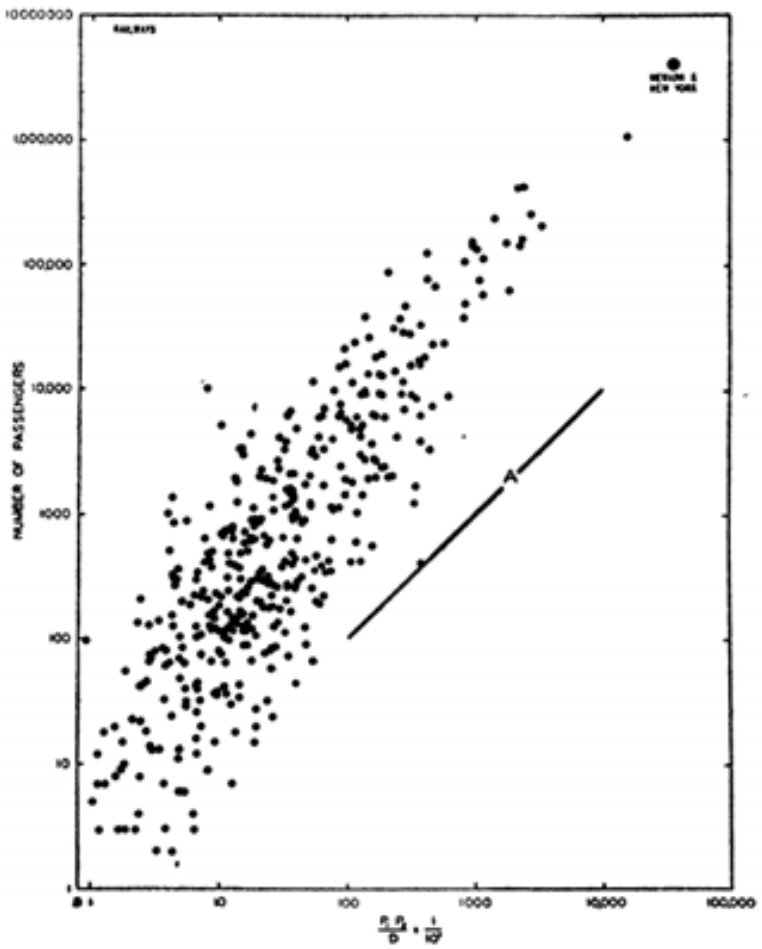
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In Figure 1 are presented the United States urban data for 1930 and 1940, as indicated, to which an ideal line, $A$, with a negative slope of 1 (i.e. $p = 1$) has been added to aid the reader's eye. The linearity of the data is apparent.

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# The generalized Gravity model

To be able to calibrate the Gravity model to different geographical and socio-economical contexts, it is often specified as:

$T_{ij}^{gravity} = A\ P_i^{\alpha}\ P_j^{\beta} / d^{\gamma}$

Our specification yesterday had these parameters
$\alpha=1$, $\beta=1$, $\gamma=2$

Where $A = \sum_i \sum_j T_{ij}^{empirical} / (P_i P_j / d^2)$

# Intervening opportunities model

*American*
SOCIOLOGICAL REVIEW

Volume 5 DECEMBER, 1940 Number 6

## INTERVENING OPPORTUNITIES: A THEORY RELATING MOBILITY AND DISTANCE*

SAMUEL A. STOUFFER
*University of Chicago*

THE movement of people in space is a basic subject of sociological inquiry. Since the classic work of Ravenstein a half century ago, numerous studies have demonstrated a close relationship between mobility and distance. Most people go a short distance; few people go a long distance.[1]

Distance is such an important factor that it needs more explicit study than it has received. Whether one is seeking to explain "why" persons go to a particular place to get jobs, "why" they go to trade at a particular store, "why" they go to a particular neighborhood to commit crime, or "why" they marry the particular spouses they choose, the factor of spatial distance is of obvious significance.

The theory here proposed and studied empirically assumes that there is no necessary relationship between mobility and distance. Instead, it introduces the concept of *intervening opportunities*. It proposes that *the number of persons going a given distance is directly proportional to the number of opportunities at that distance and inversely proportional to the number of intervening opportunities*. Another way of stating the same hypothesis is that the number of persons going a given distance is directly proportional to the percentage increase in opportunities at that distance.

# Intervening opportunities model

In other words…

*Distance is not important per se. What is important is how far one needs to travel in order to find an opportunity that suits.*

American
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# The Radiation model

# A universal model for mobility and migration patterns

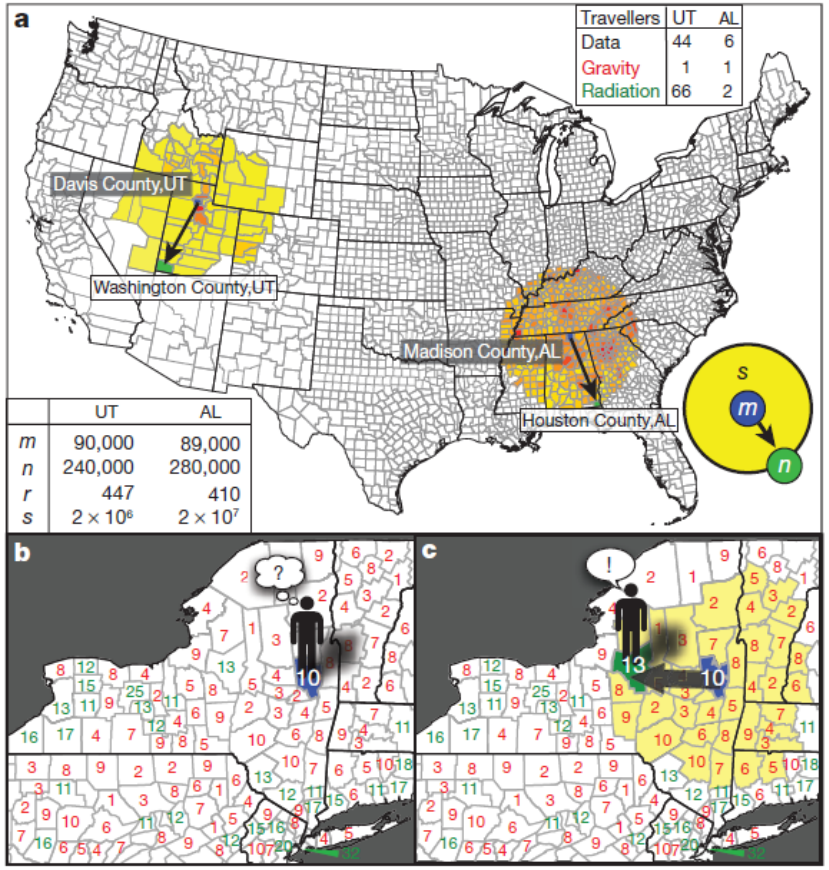
Filippo Simini[1,2,3], Marta C. González[4], Amos Maritan[2] & Albert-László Barabási[1,5,6]

**Introduced in its contemporary form in 1946 (ref. 1), but with roots that go back to the eighteenth century[2], the gravity law[1,3,4] is the prevailing framework with which to predict population movement[3,5,6], cargo shipping volume[7] and inter-city phone calls[8,9], as well as bilateral trade flows between nations[10]. Despite its widespread use, it relies on adjustable parameters that vary from region to region and suffers from known analytic inconsistencies. Here we introduce a stochastic process capturing local mobility decisions that helps us analytically derive commuting and mobility fluxes that require as input only information on the population distribution. The resulting radiation model predicts mobility patterns in good agreement with mobility and transport patterns observed in a wide range of phenomena, from long-term migration patterns to communication volume between different regions. Given its parameter-free nature, the model can be applied in areas where we lack previous mobility measurements, significantly improving the predictive accuracy of most of the phenomena affected by mobility and transport processes[11–23].**

In analogy with Newton's law of gravity, the gravity law assumes that the number of individuals $T_{ij}$ that move between locations $i$ and $j$ per unit time is proportional to some power of the population of the source ($m_i$) and destination ($n_j$) locations, and decays with the distance $r_{ij}$ between them as

$$T_{ij} = \frac{m_i^{\alpha} n_j^{\beta}}{f(r_{ij})} \qquad (1)$$

where $\alpha$ and $\beta$ are adjustable exponents and the deterrence function $f(r_{ij})$ is chosen to fit the empirical data. Occasionally $T_{ij}$ is interpreted as the probability rate of individuals travelling from $i$ to $j$, or an effective coupling between the two locations[24]. Despite its widespread use, the gravity law has notable limitations:



# Model definitions

- Generalised Gravity model

$$T_{ij}^{gravity} = A\, P_i^{\alpha}\, P_j^{\beta} / d^{\gamma}$$

- Intervening opportunities model

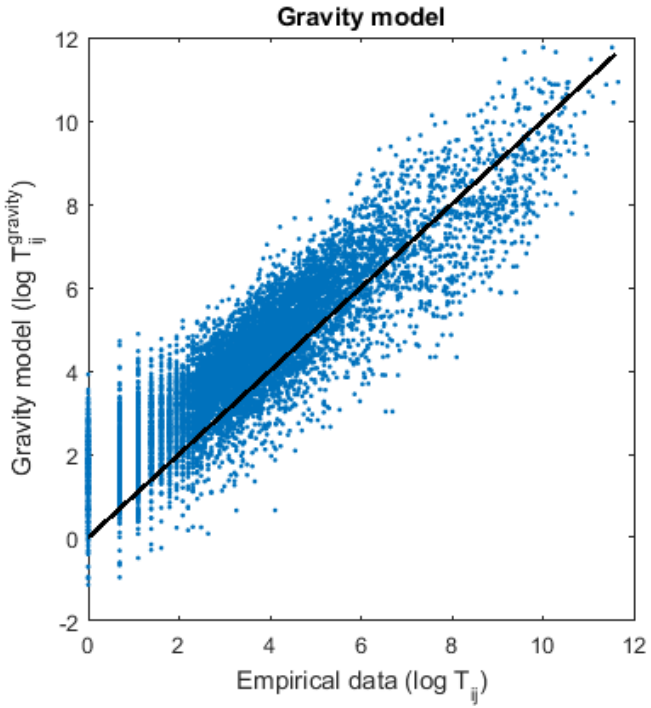
$$T_{ij}^{io} = (1-\lambda)^{(Pi + s_{ij})} - (1-\lambda)^{(Pi + s_{ij} + Pj)}$$

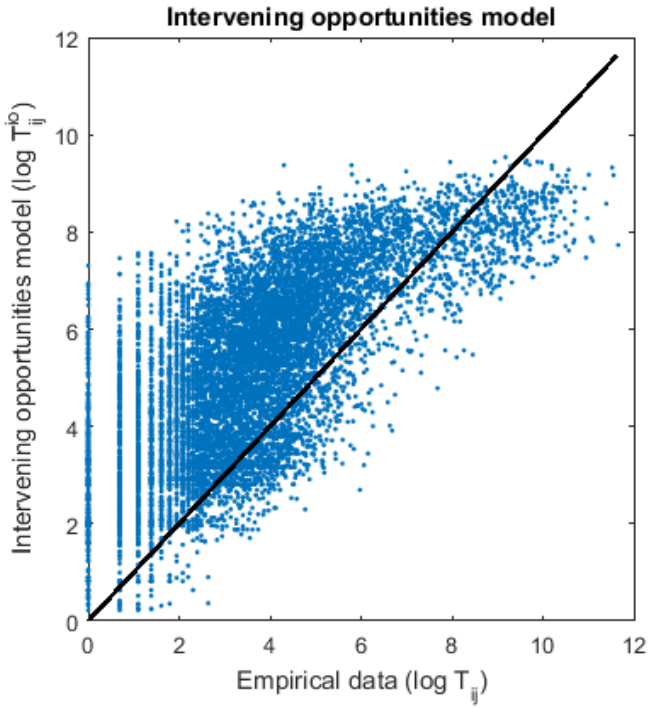
- Radiation model

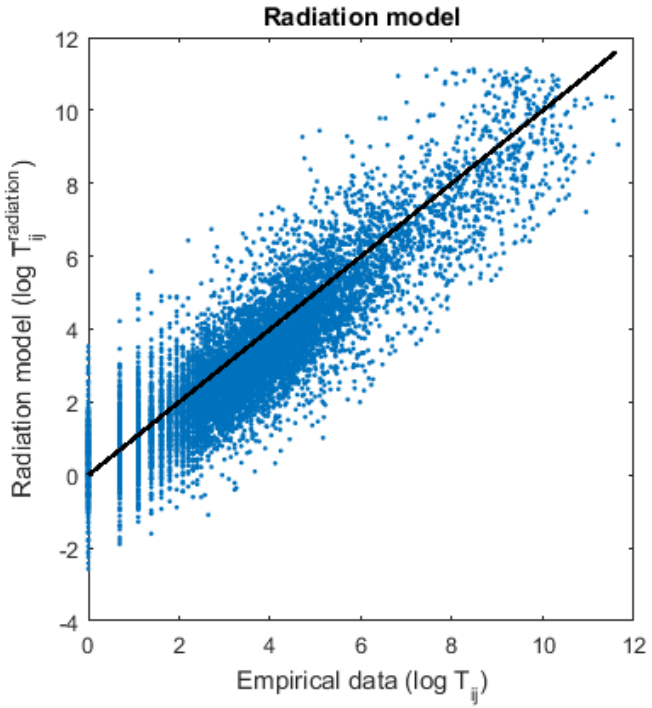
$$T_{ij}^{radiation} = T_i \frac{P_i P_j}{(Pi + sij)(Pi + Pj + sij)}$$

# Comparing models

- Let us now compare the different models on our UK Census commute dataset.



# Measure the 'goodness of fit'

Given model data $T_{ij}^{model}$ and empirical data $T_{ij}^{empirical}$, how do we measure the difference between the two data sets, and how do we then use that to compare the relative 'goodness of fit' of different models?

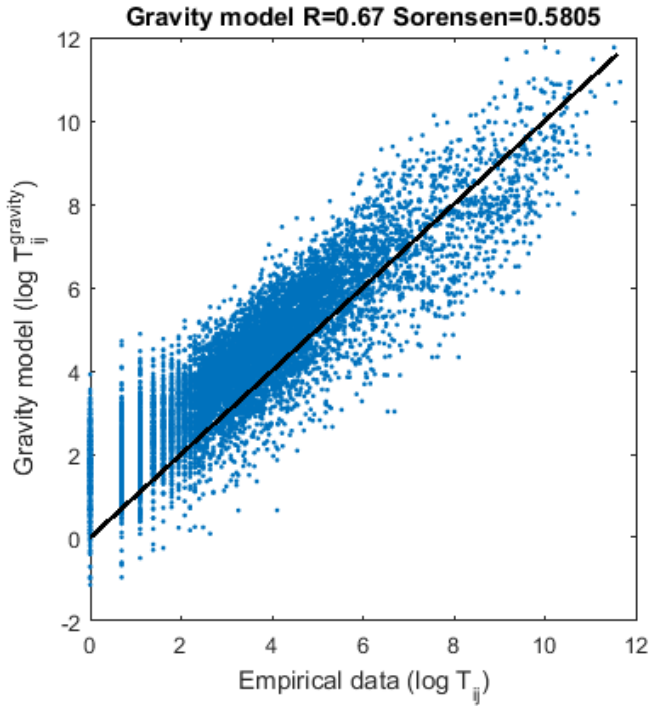
There are different measures we could use, e.g.:

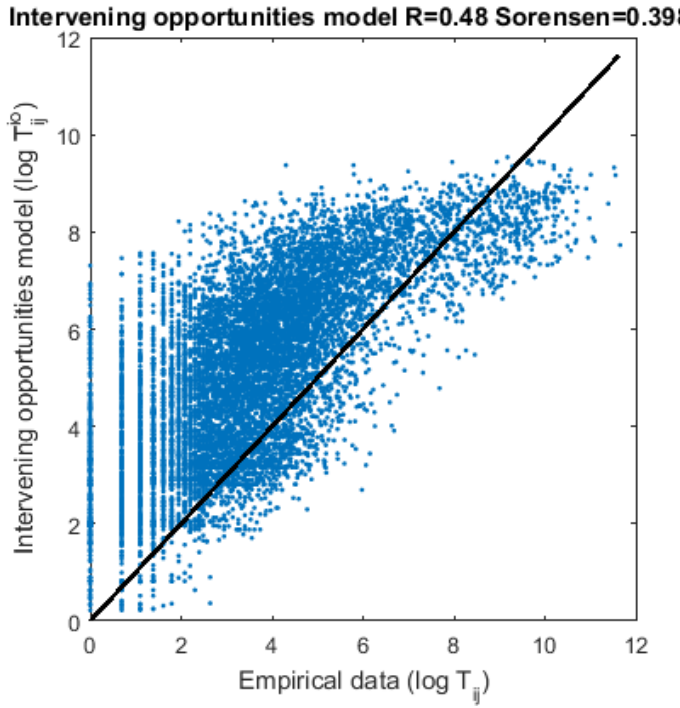
- Correlation coefficient $R = \frac{cov(T_{ij}^{model}, T_{ij}^{empirical})}{stdev\,(T_{ij}^{model})\ x\ stdev(T_{ij}^{empirical})}$

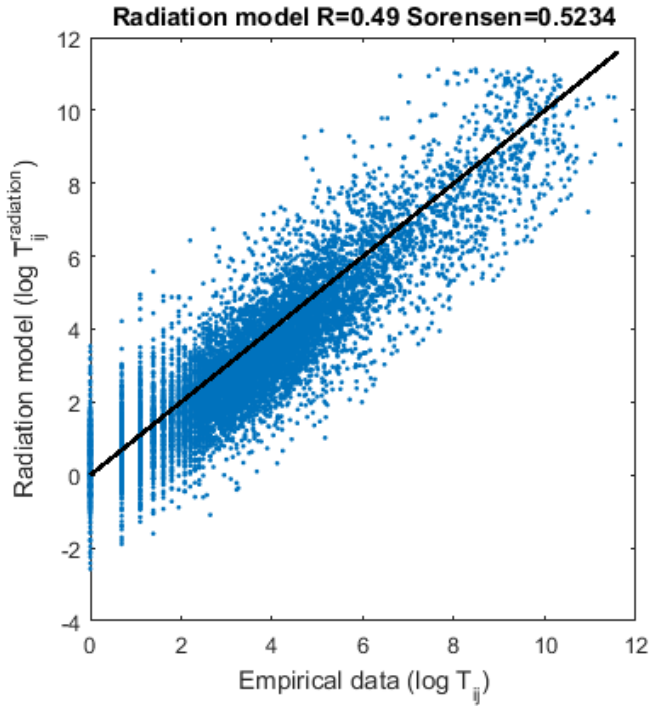
- Sorensen index $S = \frac{2\sum_{i,j}\min(T_{ij}^{model}, T_{ij}^{empirical})}{\sum_{i,j} T_{ij}^{model} + \sum_{i,j} T_{ij}^{empirical}}$

# Comparing models

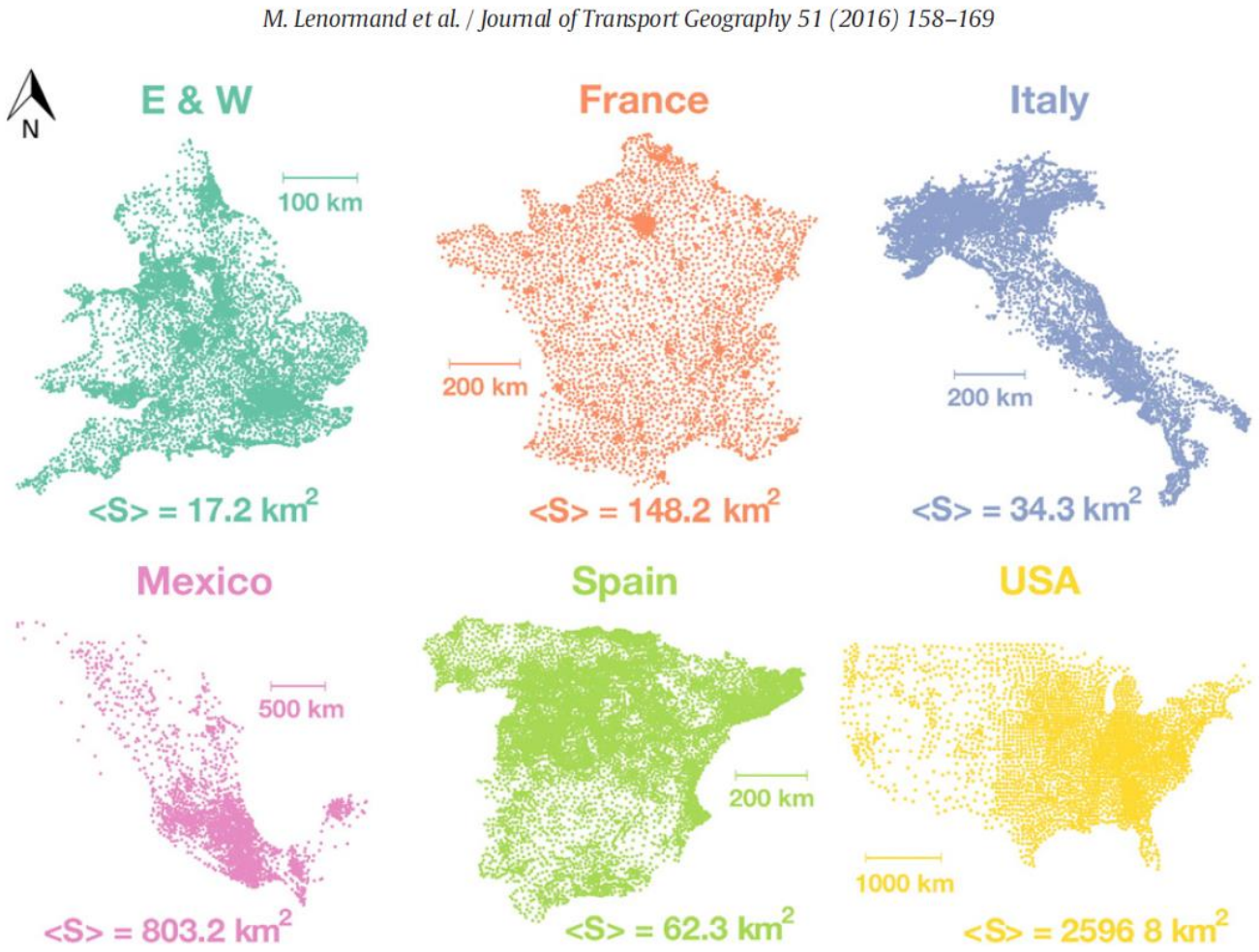
- Let us now compare the different models on our UK Census commute dataset.

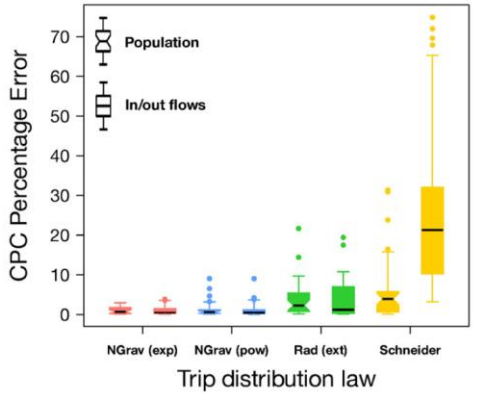


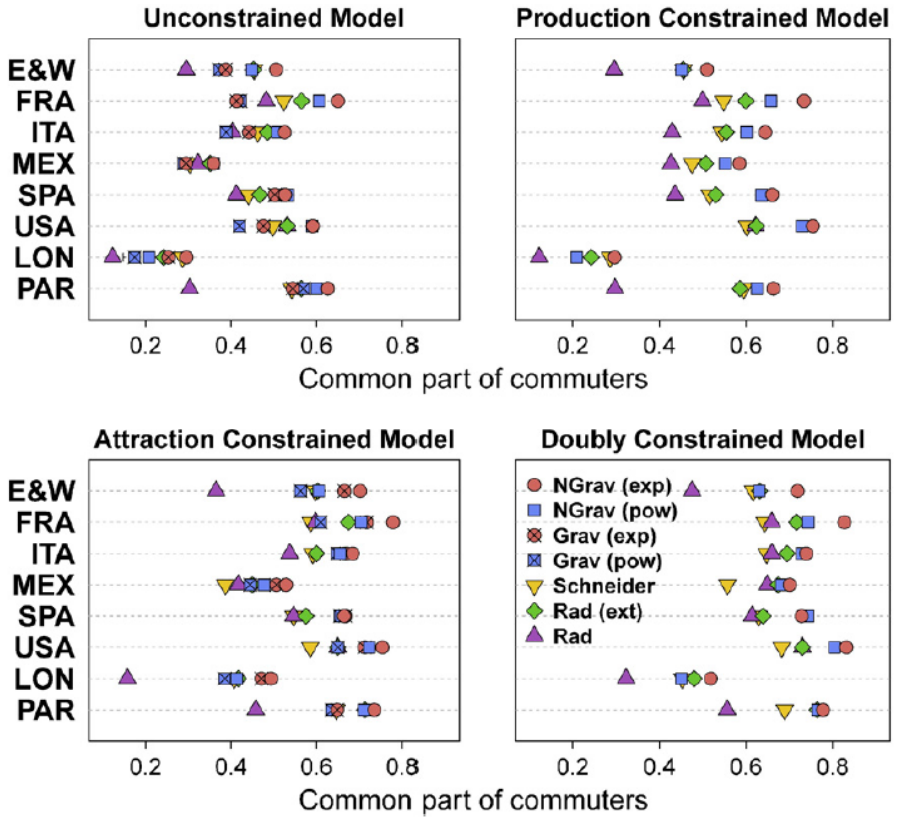



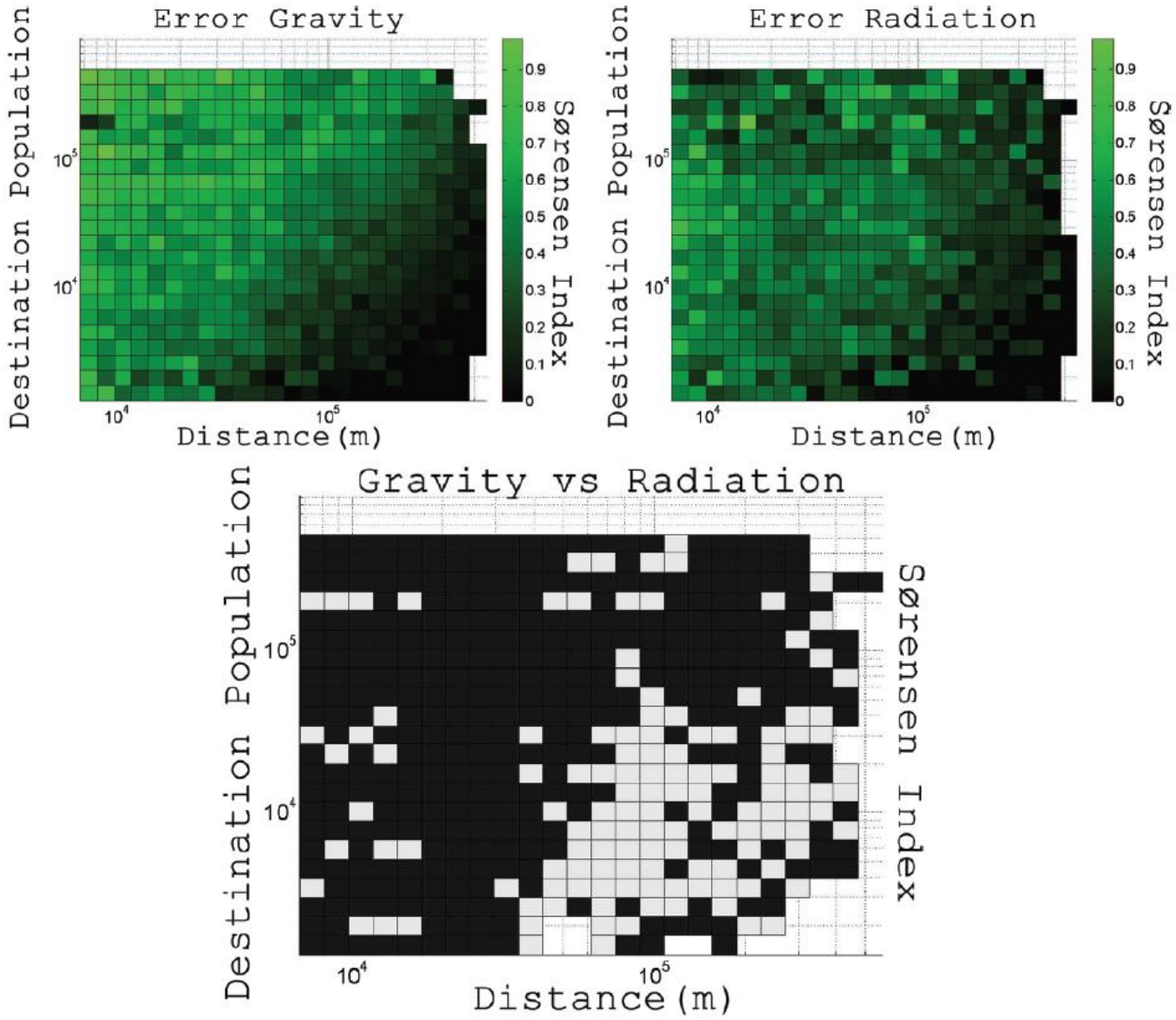
# Comparing models



# Comparing models



# What type of model is the best one?

- Simple models (few parameters) vs complicated models (lots of parameters to calibrate)
- Ad-hoc models (cf. machine learning / AI) vs models derived from a sound theoretical underpinning
- Models that requires little or no data to work vs models that requires an awful lot of data in order to work
- Models based on observable parameters vs models based on ad-hoc parameters
- Universality of results: Only works in UK but not elsewhere?
- Scale free: Works only at certain scales but not at others?

# 9. Activity based modelling

# Modelling the movement of nurses in a neonatal intensive care unit

**Background:**

- The operators of a neonatal intensive care unit want to better understand the spatio-temporal space utilisation.
- Environments where people arrive at pre-determined origins according to an arrival time distribution, and then walk towards pre-determined destinations, can be modelled with existing tools.

**The problem:**

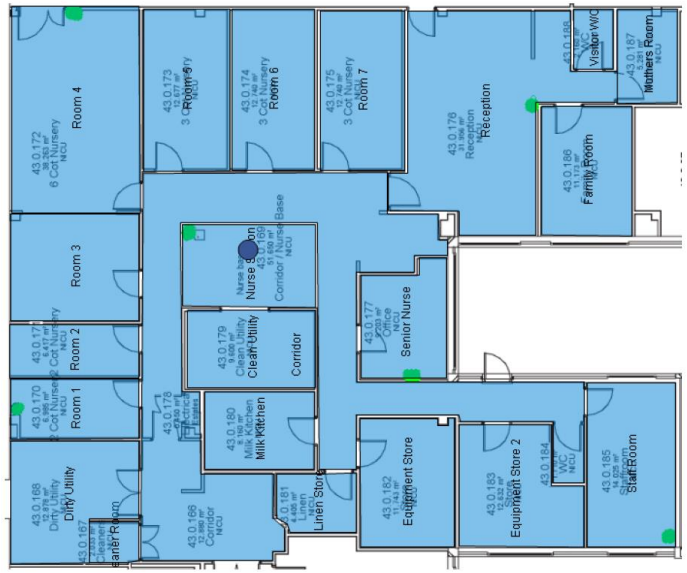
- However, the circulatory nature of staff mobility in hospital wards can not easily be modelled using existing tools, so a new approach is needed.

# Modelling the movement of nurses in a neonatal intensive care unit

- Nurses were tracked by RFID/WiFi for a month to reveal their movement patterns. But what is the process that *generated* those trips?

# Modelling the movement of nurses in a neonatal intensive care unit

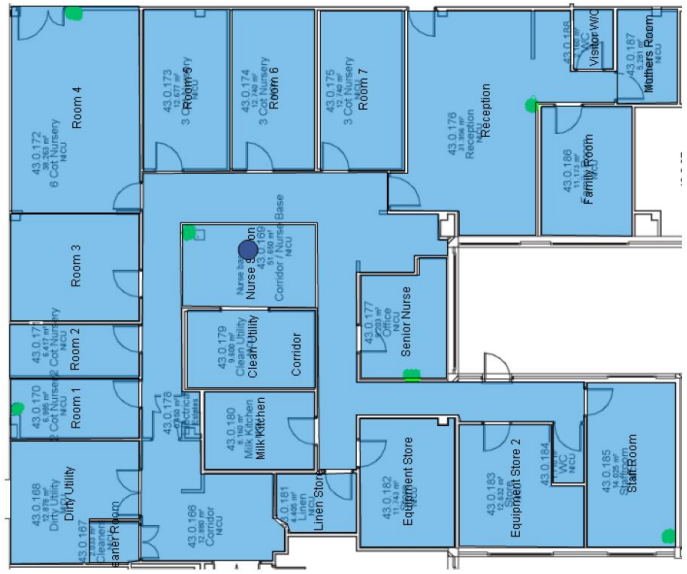
- Nurses were tracked by RFID/WiFi for a month to reveal their movement patterns. But what is the process that *generated* those trips?




Layout of Case Study NICU Department

# Modelling the movement of nurses in a neonatal intensive care unit

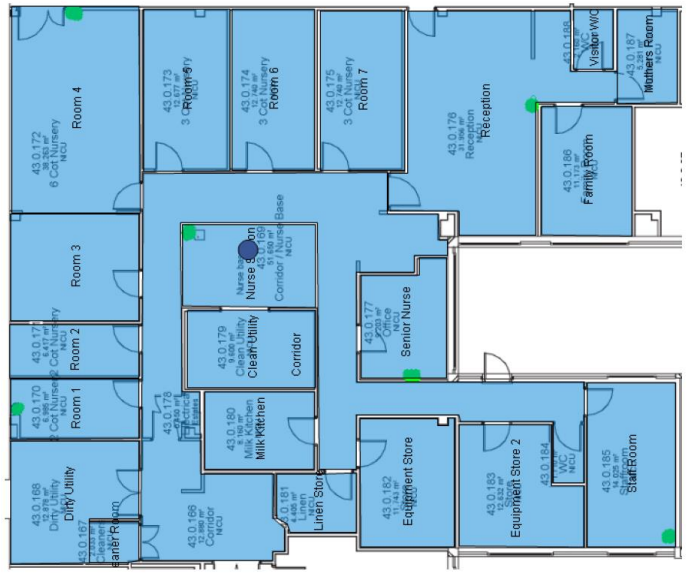
- Null model (Model 1): Random room assignment.
- Accuracy: ~5%




Layout of Case Study NICU Department

# Modelling the movement of nurses in a neonatal intensive care unit

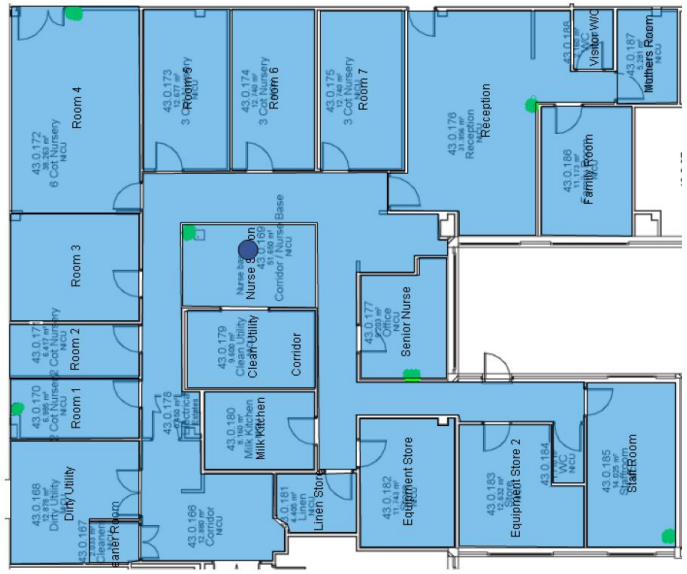
- Model 2: Attractiveness (assuming some rooms are more attractive than others); and (3) Temporal Attractiveness (the same approach but a different model for each hour). Accuracy: ~17% in both cases.



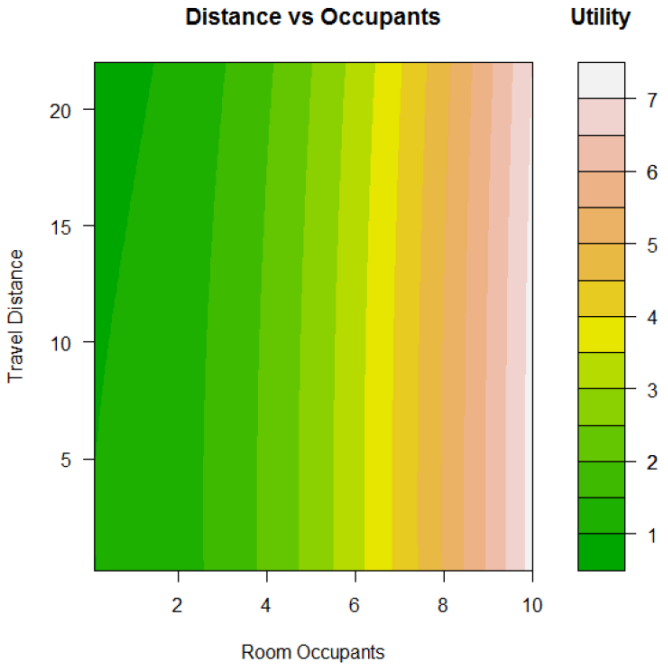

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# Modelling the movement of nurses in a neonatal intensive care unit

- Model 4: Occupancy-Distance model (to capture social interaction).
- Accuracy: ~31%

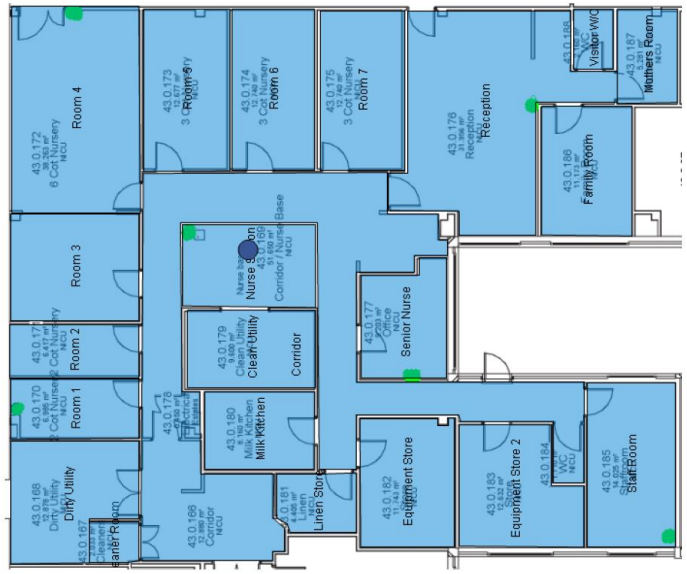



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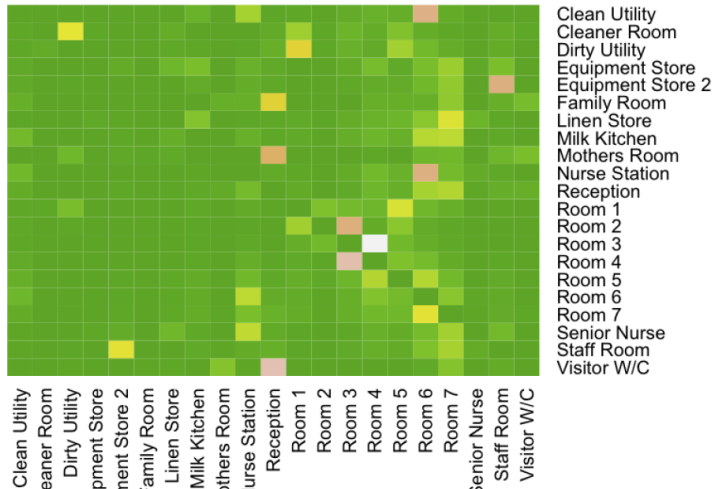


# Modelling the movement of nurses in a neonatal intensive care unit

- Model 5: Markov Chain Model.
- Accuracy: ~39%

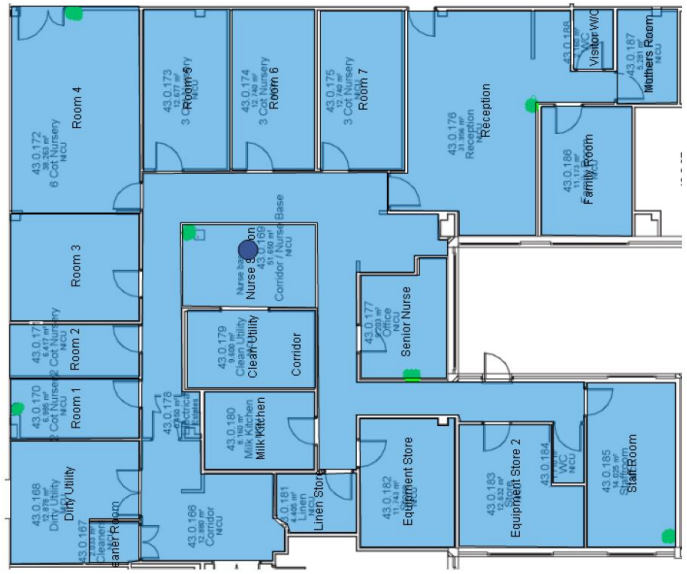



Layout of Case Study NICU Department

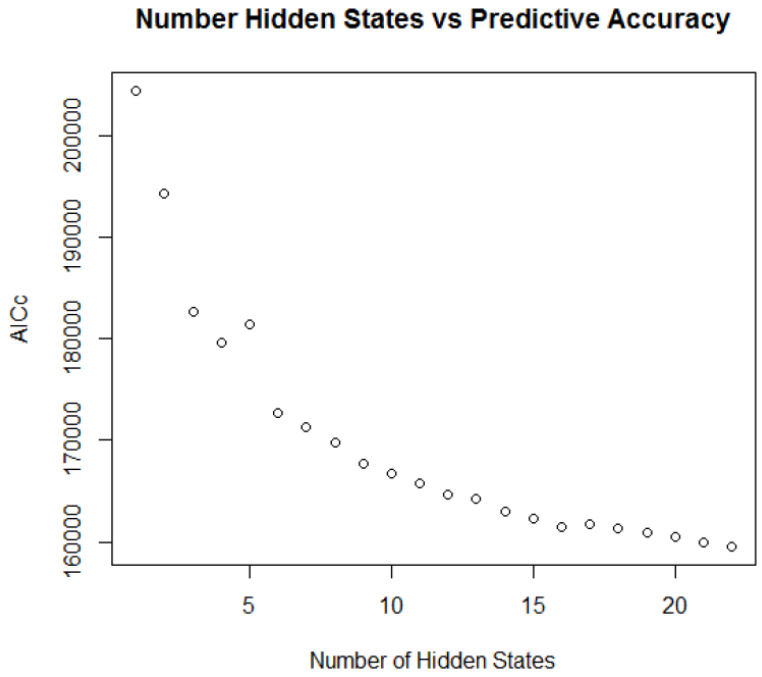


# Modelling the movement of nurses in a neonatal intensive care unit

- Model 6: Hidden Markov Model (22 hidden states).
- Accuracy: ~88%




Layout of Case Study NICU Department

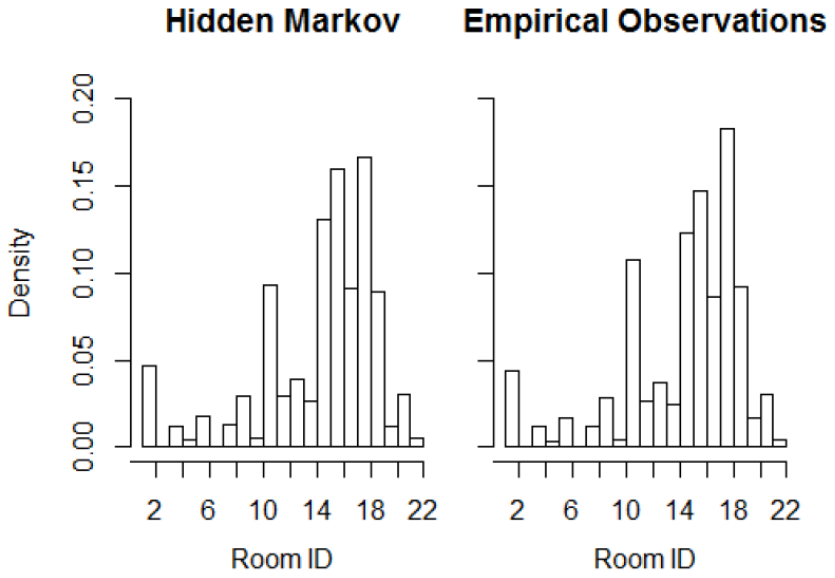


# Modelling the movement of nurses in a neonatal intensive care unit

- Different types of room transition models are compared, to see which one can best explain the empirical data.

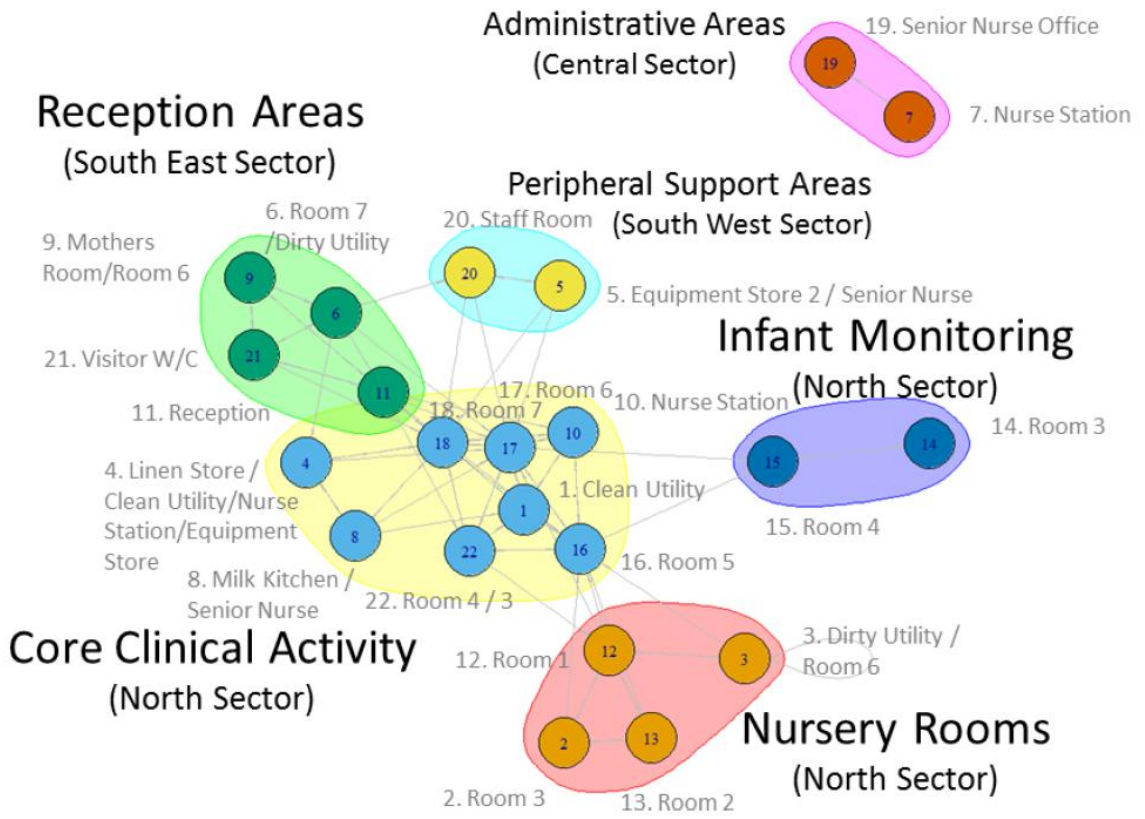
| Model Name | % Next Room | Improvement |
|---|---|---|
| Random | ~5% | Baseline |
| Temporal Attractiveness | ~17% | 3.2 x |
| Attractiveness | ~17% | 3.2 x |
| Occupancy Distance | ~31% | 6.2x |
| Markov Chain | ~39% | 7.8x |
| Hidden Markov (22 States) | ~88% | 17.6x |

Goodness-of-Fit of Transition Models Using Full Dataset



# Modelling the movement of nurses in a neonatal intensive care unit

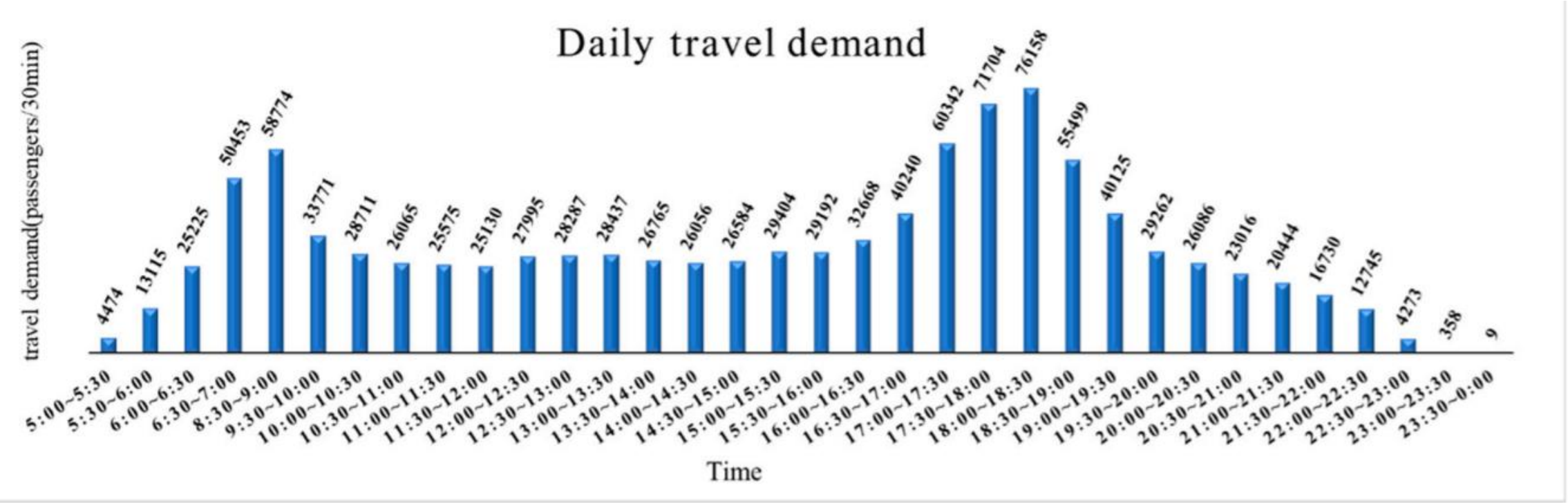
- Clusters identified by the Hidden Markov Model.




Characteristic Diagram of NICU Nurse HMM Room Transition Model. The characteristic diagram represents the clusters of activity as identified by the HMM. For the purposes of visualisation each hidden state is mapped to the most probable room thus enabling a spatial representation of the hidden states and their interconnections. For the purposes of clustering the hidden states the walk-trap algorithm is used to highly which hidden states are tightly associated.

# 10. Temporal behavior of demand

# Beijing Metro demand

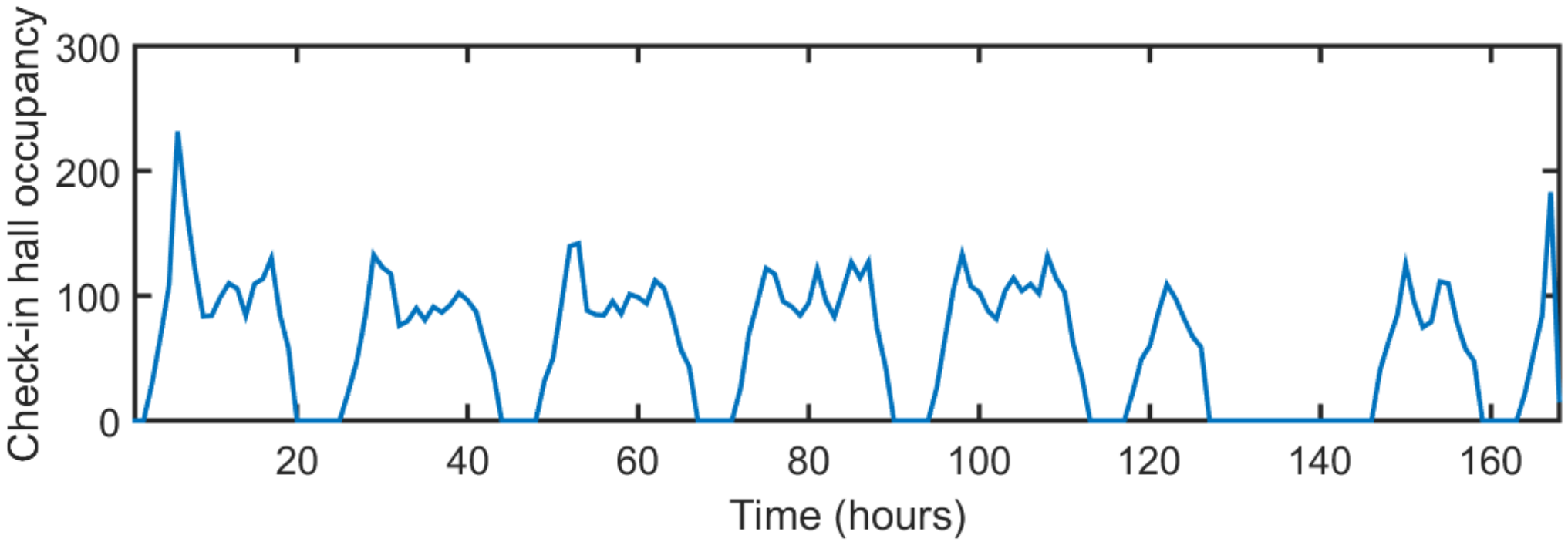



Daily travel demand of Beijing Metro line 4.

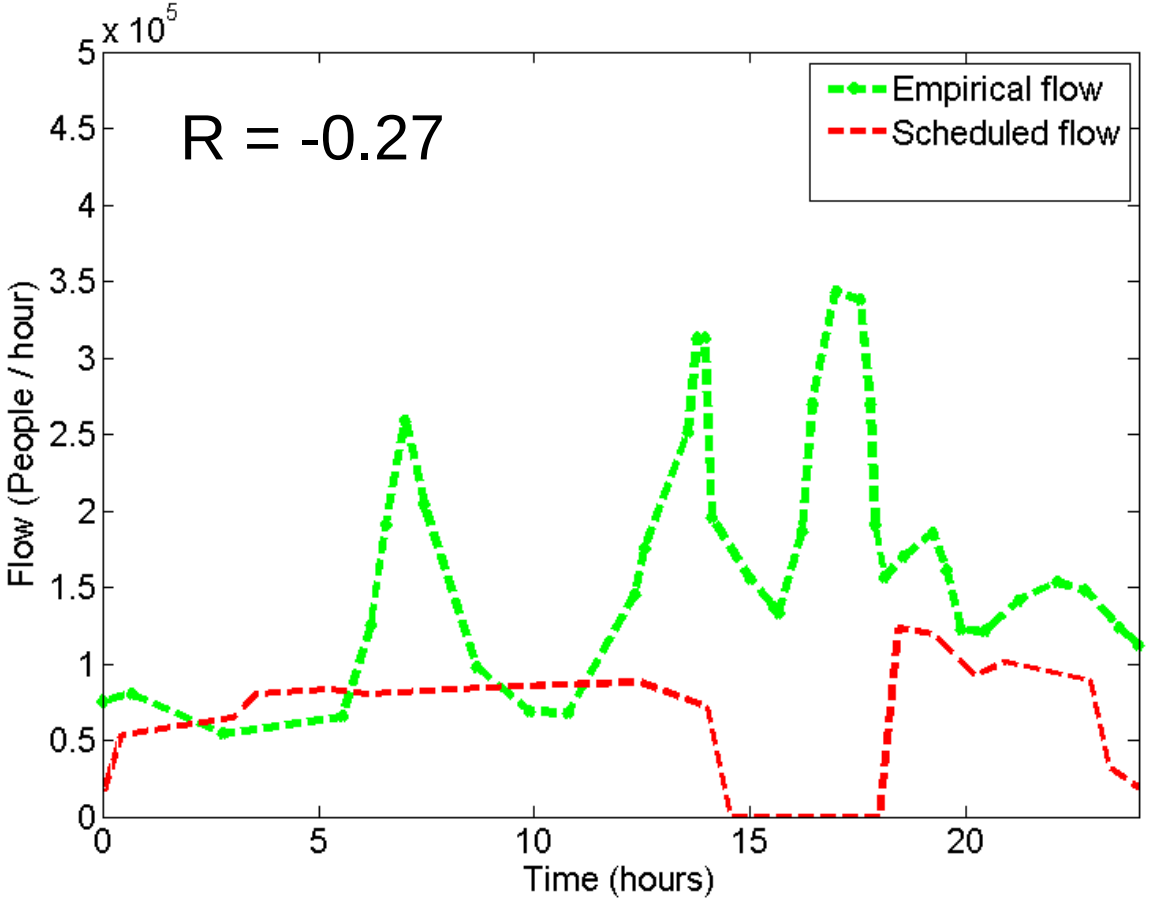
Yin et al (2019)
Sustainability 11(13), 3701

# Airport demand

Occupancy in an airport check-in hall over 1 week.



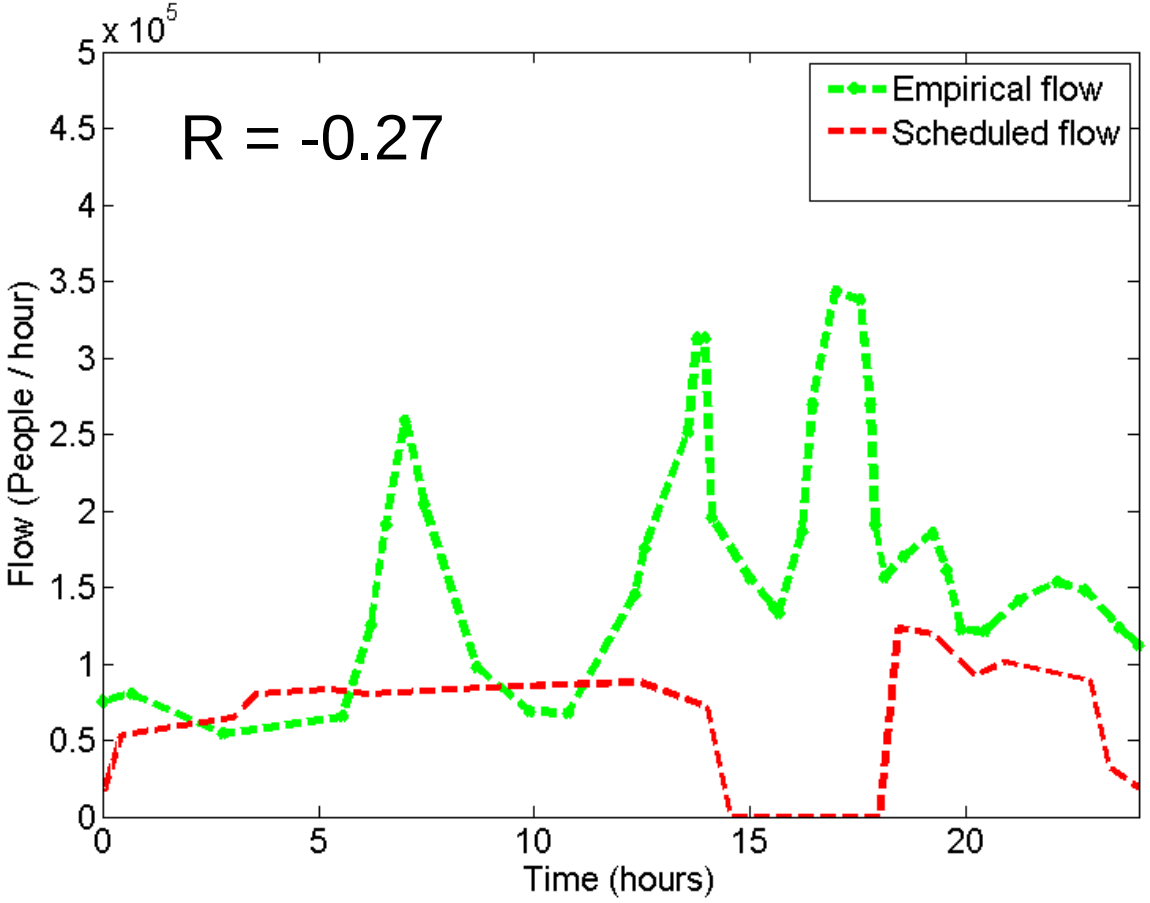
# Crowd flow behaviour at the Hajj: Turn-up profile over a day




How people really behave

How people are told to behave (schedules)

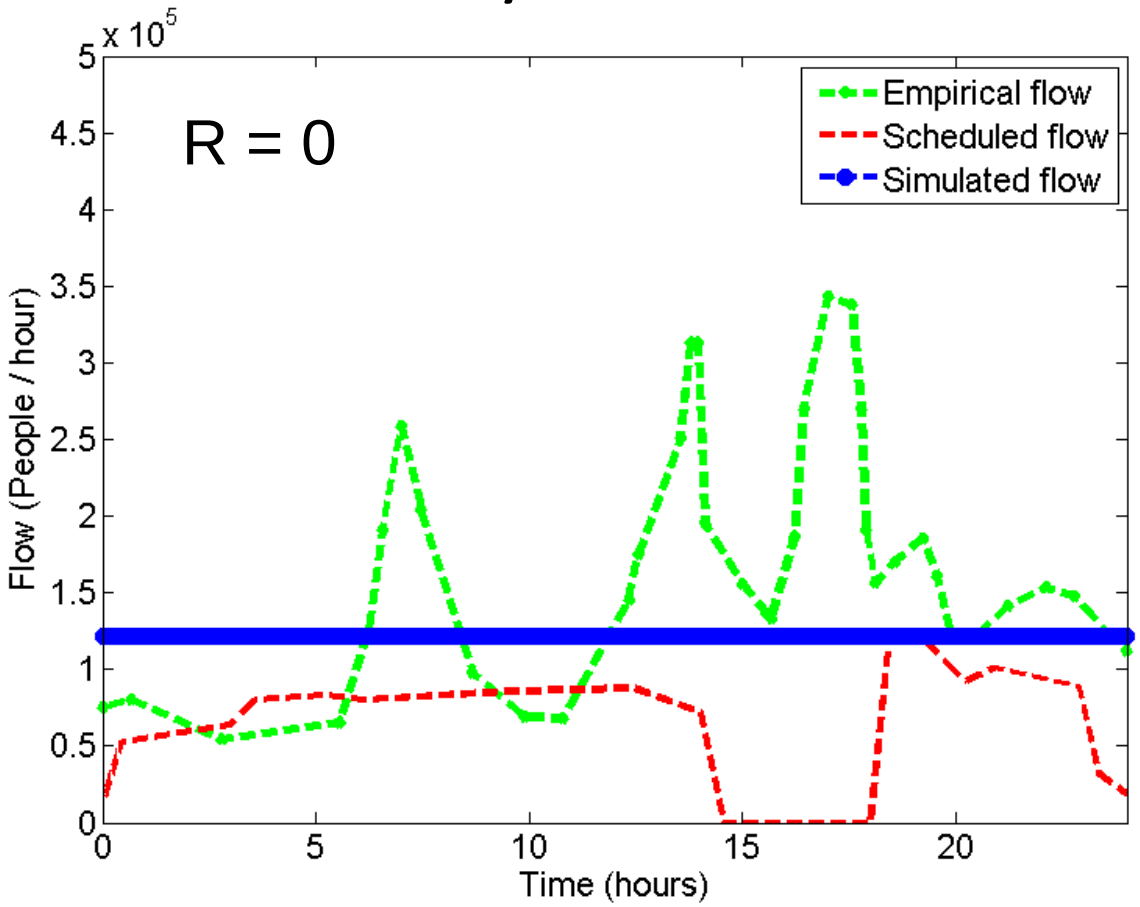
# Crowd flow behaviour at the Hajj: Turn-up profile over a day




How people really behave

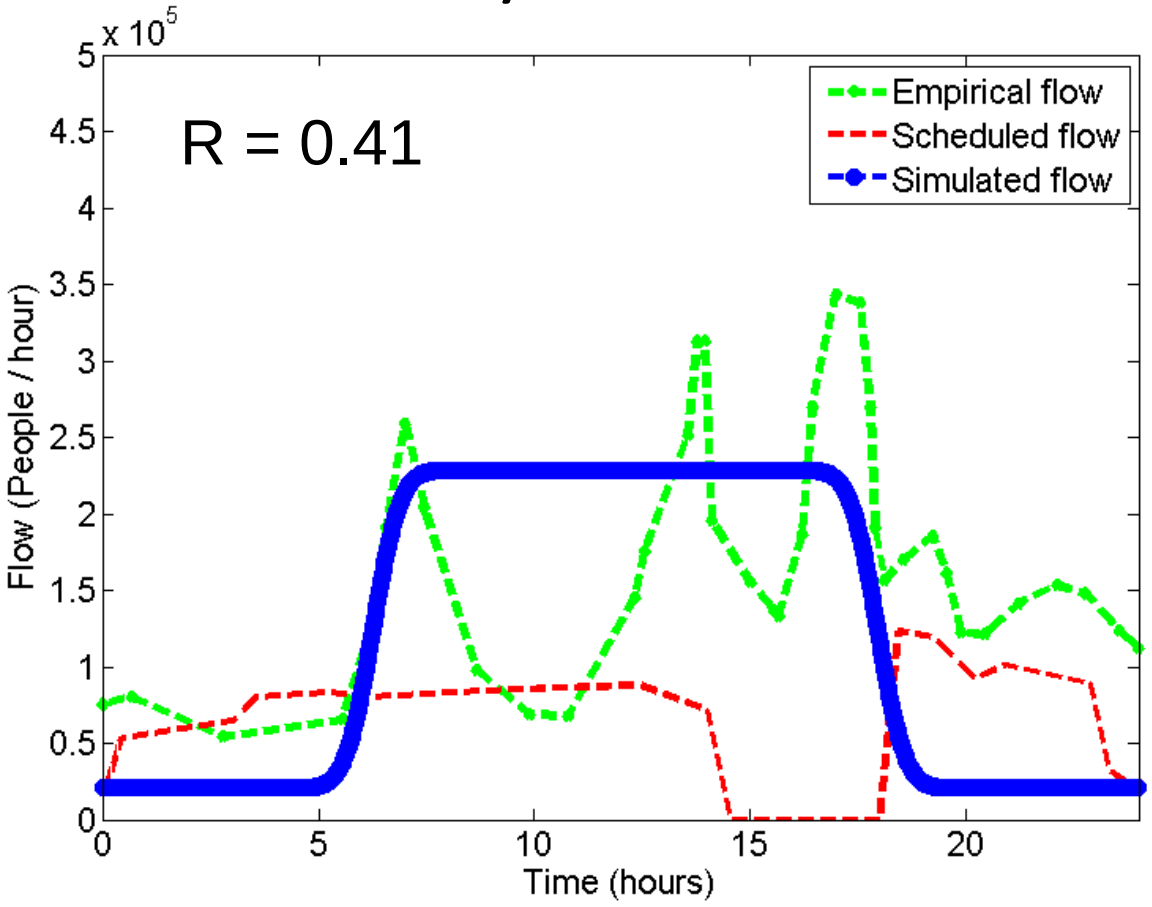
How people are told to behave (schedules)

# Crowd flow behaviour at the Hajj: Turn-up profile over a day



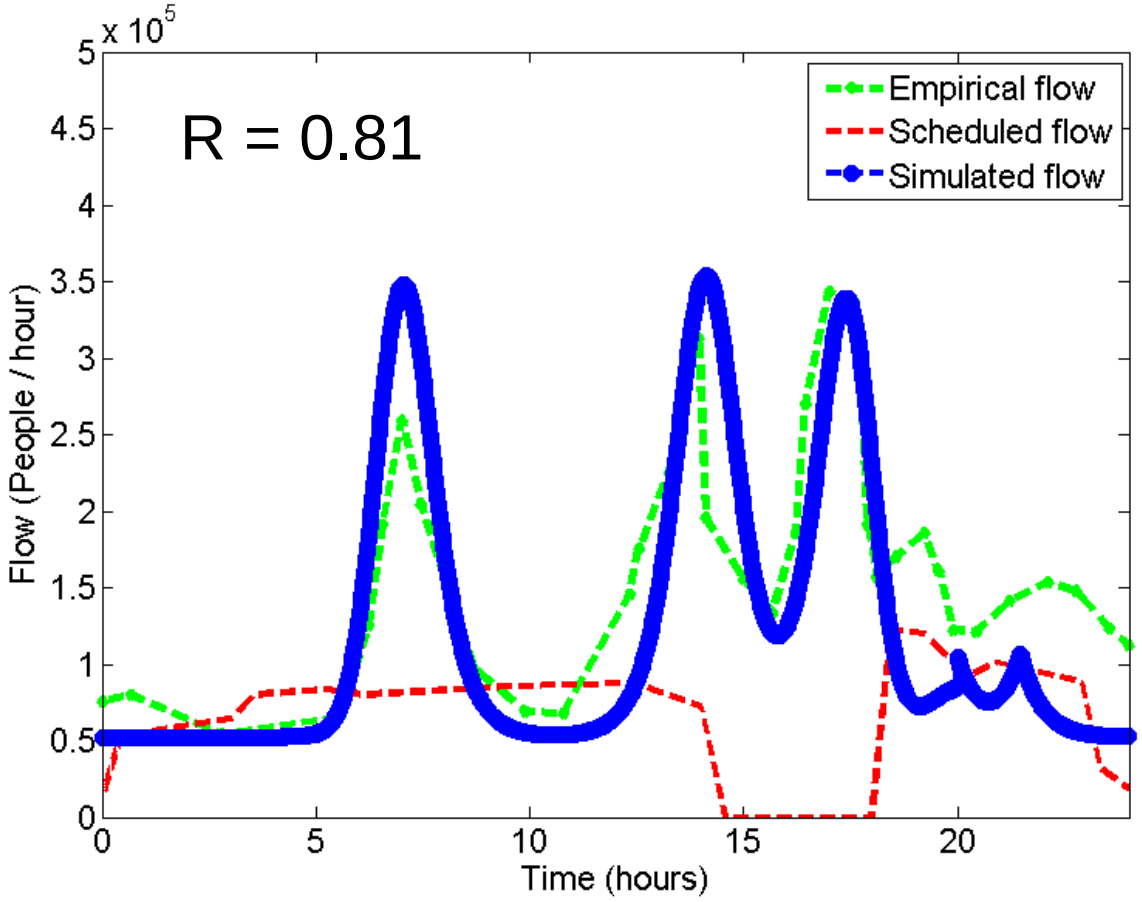

Model = null model

# Crowd flow behaviour at the Hajj: Turn-up profile over a day




Model = daylight model

# Crowd flow behaviour at the Hajj: Turn-up profile over a day




**Makkah, Saudi Arabia Prayer Times**

**Prayer Schedule October / 2014**

| Date | Day | Fajr | Sunrise | Dhuhr | Asr | Maghrib | Isha |
|---|---|---|---|---|---|---|---|
| 15 | Wed | 5:01 | 6:16 | 12:07 | 3:27 | 5:57 | 7:27 |
| 16 | Thu | 5:01 | 6:16 | 12:07 | 3:27 | 5:56 | 7:26 |
| 17 | Fri | 5:01 | 6:17 | 12:07 | 3:26 | 5:55 | 7:25 |
| 18 | Sat | 5:02 | 6:17 | 12:06 | 3:26 | 5:55 | 7:25 |
| 19 | Sun | 5:02 | 6:17 | 12:06 | 3:26 | 5:54 | 7:24 |
| 20 | Mon | 5:02 | 6:18 | 12:06 | 3:25 | 5:53 | 7:23 |
| 21 | Tue | 5:03 | 6:18 | 12:06 | 3:25 | 5:52 | 7:22 |

Monthly/Annual Schedule

Model = daylight + praying model